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Master's Programme in Information and Service Management

Comparing Predictive Methods for Customer Churn

A Case Study of a B2B SaaS Company

Eino Rask

Master's thesis
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Abstract

Customer churn is a significant challenge for many Business-to-Business (B2B) Software-as-a-Service (SaaS) companies, which often depend on recurring revenue. While companies collect large amounts of data, they often lack a clear understanding of the key factors that lead to customer churn. Therefore, it is important for these companies to use data-driven methods to predict churn and to understand its underlying causes.

This study uses a quantitative method to investigate customer churn prediction within a case study of a B2B SaaS company operating in the Nordic. Several machine learning models were implemented and compared, including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, and XGBoost. The dataset was extracted from the case company's CRM system and included customer characteristics, financial data, and internally developed Customer Health Score metrics. Hyperparameter tuning was used to find the optimal settings for each model. The models were then trained and evaluated. Finally, model interpretation techniques, specifically Permutation Feature Importance (PFI) and SHAP analysis, were applied to identify the most important churn drivers.

The results conclude that the ensemble-based models, particularly XGBoost, performed better than the other algorithms. The XGBoost model achieved the highest performance across key metrics, including an exceptionally high recall score (1.0) on the test data. The analysis of churn drivers revealed that "relation_strength", a composite metric measuring the overall customer relationship and engagement, was the single most important predictor. The main conclusion of this thesis is that the quality of the customer relationship is a more critical driver of retention than individual usage or business metrics in this B2B SaaS context.

Keywords Customer Churn, Churn Prediction, Churn Drivers, Machine Learning, Customer Relationship Management, Software-as-a-Service, B2B

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Tiivistelmä

Asiakaspoistuma on merkittävä haaste monille Business-to-Business (B2B) Software-as-a-Service (SaaS) -yrityksille, joille toistuva tulovirta on usein keskeinen tulonlähde. Yrityksiltä puuttuu kuitenkin usein selkeä ymmärrys keskeisistä tekijöistä, jotka johtavat asiakaspoistumaan. Tämän vuoksi yritysten on tärkeää hyödyntää dataohjautuvia menetelmiä poistuman ennustamiseksi ja sen taustasyiden ymmärtämiseksi.

Tässä tutkimuksessa tarkastellaan asiakaspoistuman ennustamista kvantitatiivisella menetelmällä Pohjoismaisessa B2B SaaS -yrityksessä toteutetussa tapaustutkimuksessa. Työssä toteutettiin ja vertailtiin useita koneoppimis malleja, mukaan lukien Logistic Regression, Support Vector Machine, Decision Tree, Random Forest ja XGBoost. Aineisto kerättiin kohdeyrityksen CRM-järjestelmästä, ja se sisälsi asiakkaiden ominaisuustietoja, taloudellista dataa sekä yrityksen sisäisesti kehittämiä Customer Health Score -mittareita. Jokaiselle mallille etsittiin optimaaliset asetukset hyperparametrien virityksellä, minkä jälkeen mallit koulutettiin ja arvioitiin. Lopuksi parhaiten suoriutuneisiin malleihin sovellettiin tulkintamenetelmiä, kuten Permutation Feature Importance (PFI) ja SHAP-analyysia, tärkeimpien muuttujien tunnistamiseksi.

Tulokset osoittivat, että ensemble-pohjaiset mallit, erityisesti XGBoost, suoriutuivat muita algoritmeja paremmin. XGBoost-malli saavutti parhaan suorituskyvyn keskeisillä mittareilla, mukaan lukien poikkeuksellisen korkean recall-arvon (1.0) testidatalla. Muuttujien analyysi paljasti, että ”relation_strength”, joka on asiakassuhteen kokonaisvaltaista vahvuutta ja sitoutuneisuutta mittaava yhdistelmämittari, oli tärkein yksittäinen ennustaja. Tämän tutkielman tärkein johtopäätös on, että asiakassuhteen laatu on tässä B2B SaaS -kontekstissa merkitsevämpi muuttuja kuin yksittäiset käyttöön tai liiketoimintaan liittyvät mittarit.

Avainsanat Asiakaspoistuma, Asiakaspoistuman ennustaminen, Asiakaspoistuman ajurit, Koneoppiminen, Asiakkuudenhallinta, Software-as-a-Service, B2B

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Preface and acknowledgements

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1 Introduction

1.1 Background and motivation

In today's competitive business world, keeping existing customers is crucial for company success and long-term earnings. Acquiring new customers is often much more expensive than retaining current ones (Stucki, 2019) Because of this, customer churn, which means losing customers, presents a major challenge for many businesses. Understanding and reducing churn has therefore become a key goal for companies aiming for steady growth.

This challenge is particularly pronounced in the Software-as-a-Service (SaaS) industry. The SaaS business model relies on recurring subscription fees. This makes customer retention and minimizing churn essential for sustained profitability and growth (Ge et al., 2017). The financial impact of churn in the SaaS sector is significant. SaaS businesses often face high customer acquisition costs (CAC). These costs include marketing, sales commissions, and initial customer onboarding. According to Ge et al. (2017), it can take an average of twelve months for a SaaS company to recover the costs of acquiring a single new customer. Figure 1 visually illustrates this extended payback period, showing the initial negative cash flow before a customer starts generating profit.

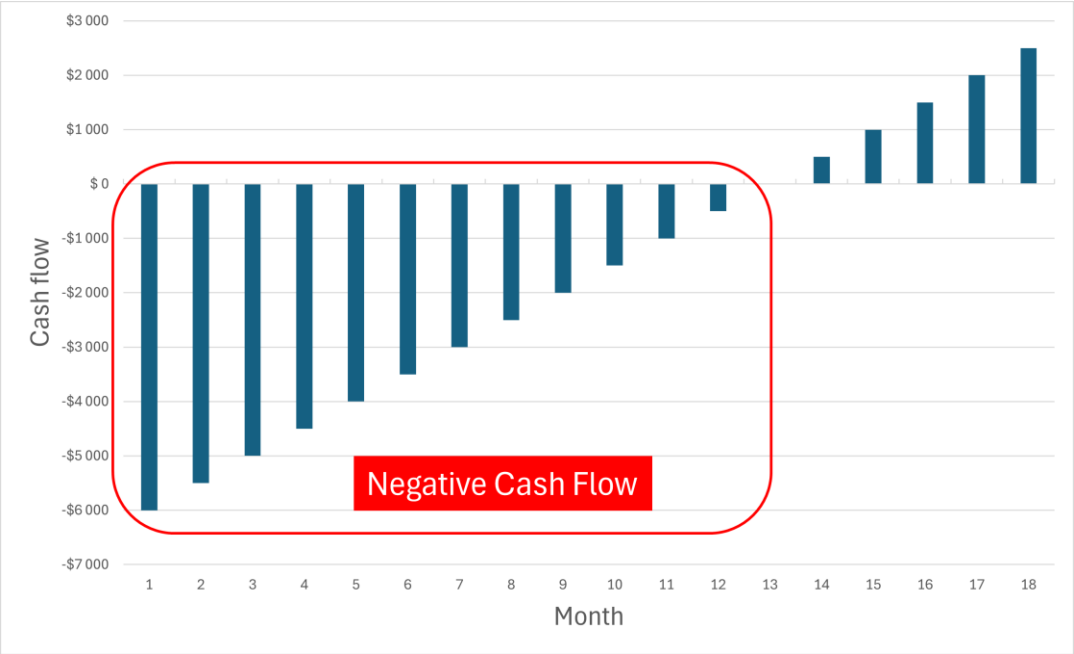

Figure 1: Illustrative single customer cumulative cash flow in a SaaS Model (Ge et al., 2017)

As a result, early customer churn directly harms profitability. If a customer leaves before this break-even point, often before the 13th month, the company does not fully recover its investment. This leads to a net loss for that

specific customer (Ge et al., 2017). The sooner a customer departs, the greater the financial loss for the SaaS provider. This highlights why customer retention is extremely critical for SaaS companies, especially in competitive markets (Ge et al., 2017). The SaaS market is expanding rapidly. However, increasing competition makes it difficult to identify the key factors for successful customer retention. These further stresses the need for effective churn analysis (Ge et al., 2017). Furthermore, understanding churn drivers helps companies take proactive steps to keep customers. It can also provide insights for more effective marketing campaigns by showing which software features or benefits are most valued (Ge et al., 2017).

The need to predict and manage churn becomes even more complex for businesses operating in a Business-to-Business (B2B) context within the SaaS industry. Predicting customer churn in B2B settings has unique challenges compared to Business-to-Customer (B2C) situations. B2B area has also received less attention in academic research (Figalist et al., 2019). Several key differences explain why B2B churn prediction is particularly demanding and makes the focus of this thesis relevant.

First, B2B companies typically serve a smaller number of customers, but each customer often represents a much higher transaction value. Losing a single B2B customer can therefore have a substantially greater financial impact than in B2C. This makes retaining every B2B client vital for the financial stability of SaaS providers (Figalist et al., 2019).

Second, B2B environments involve more complex operational settings and more stakeholders in the churn decision. Figalist et al. (2019) point out a key difference: the customer company making the purchase, is distinct from the end-users, who are usually employees using the product daily. The customer company's organization decides whether to continue or end a subscription. However, the satisfaction and experiences of these end-users can strongly influence the decision. This multi-layered relationship requires a careful approach to data in B2B churn prediction. It is potentially beneficial to combine both organizational data and end-user data like usage and engagement metrics (Figalist et al., 2019).

Third, the perceived value of a product can vary significantly between B2C and B2B customers. It can also differ among various stakeholders within a single B2B client company. For example, the purchasing department might focus on return on investment and strategic fit. In contrast, end-users might prioritize ease of use and specific features (Figalist et al., 2019). These different views on value can make retention difficult if the SaaS provider does not understand and address them. Figalist et al. (2019) therefore emphasize that B2B churn prediction models need to effectively use diverse data sources. These sources include contract details, interaction histories, usage patterns, and support experiences.

These unique aspects of the B2B market, such as higher customer value, complex stakeholder interactions, and varied value perceptions, along with

the SaaS financial model, show why B2C churn prediction methods are often not enough for B2B companies. This highlights the need for specific research on B2B churn. This thesis, for example, focuses on customer churn prediction within a specific B2B SaaS company, Lime Technologies Finland. While Lime Technologies generally has strong customer retention in many markets, recent trends in the Finnish market showed an opportunity for improvement, with an internal churn rate of about six percent in 2023. This gave a clear business reason for the company to explore how data-driven predictive methods could support their wider goal of strengthening customer loyalty. Also, this study is a quantitative follow-up to a qualitative research project on churn drivers that the case company did in 2023. The aim is to build on those first insights by developing and evaluating predictive models. By developing and comparing predictive models for this context, this study aims to offer practical insights. These insights can help the case company, and possibly other B2B SaaS firms, to better understand and reduce customer churn.

1.2 Research objectives and questions

The main goal of this thesis is to study and compare how well different predictive methods work for customer churn in a B2B SaaS context. This will be done using data from a specific case company. In addition, this study aims to find the key factors that cause churn in this specific company and to give useful recommendations for improving customer retention.
To reach this main goal, the following research questions have been set:

- RQ1: Which modeling techniques provide the most effective predictions for customer churn using the case company's data?

- RQ2: What are the key drivers of customer churn identified by the most effective predictive models in this case study?

- RQ3: What practical recommendations for customer retention can be derived from the analysis of model performance and churn drivers?

The research questions for this thesis have been formulated to address specific gaps identified in the existing literature on customer churn, particularly within the B2B SaaS context.

The first research question (RQ1: Which modeling techniques provide the most effective predictions for customer churn using the case company's data?) is posed to address a research gap concerning the comparative performance of different predictive methodologies. While numerous studies have applied machine learning to churn prediction, prior B2B SaaS research often lacks direct, head-to-head comparisons of traditional statistical models (like

logistic regression) against a diverse range of individual and ensemble machine learning classifiers using real-world, company-specific data (Dahlén & Mauritzon, 2023; Stucki, 2019). This study aims to fill this gap by providing empirical evidence on which modeling techniques achieve the most effective churn predictions in the context of the case company's B2B SaaS operations.

Secondly, RQ2 (What are the key drivers of customer churn identified by the most effective predictive models in this case study?) responds to the frequently highlighted "interpretability gap" in quantitative churn studies. While achieving high predictive accuracy is a common goal, several researchers emphasize the critical need to also understand and explain why customers churn, not just how well models can predict this outcome (Dahlén & Mauritzon, 2023; Stucki, 2019; Khare & Arora, 2024). Many existing quantitative studies do not sufficiently clarify the underlying reasons for customer attrition. By using model interpretation techniques, this study aims to identify the key drivers of churn. It will look at how the company's own Customer Health Score metrics and other specific indicators affect this, trying to give more in-depth insights that are more than just predictions. This method also tackles the need for research that includes and tests important factors that are often left out of general datasets, such as detailed information on customer engagement and perceived satisfaction (Rautio, 2019; Dahlén & Maurizon, 2023).

Finally, RQ3 (What practical recommendations for customer retention can be derived from the analysis of model performance and churn drivers?) aims to close the gap in practice-oriented recommendations. Previous studies have strongly emphasized that the true benefit of churn prediction models comes when their analytical results are turned into specific, actionable strategies for retention (Dahlén & Mauritzon, 2023; Khare & Arora, 2024). Therefore, by systematically analyzing both the performance of the predictive models and the identified key churn drivers, this study aims to develop practical, evidence-based recommendations. These recommendations are intended to help the case company enhance its customer retention efforts, thereby ensuring that the analytical insights generated are directly connected to real-world business applications and value.

1.3 Structure of the thesis

This thesis has six main chapters, which guide the reader through the research process from introduction to conclusions.

Chapter 1 gives the background and reasons for this study. It outlines the main goals and the specific research questions this thesis aims to answer. This chapter also describes the overall structure of the thesis.

Chapter 2 reviews existing academic literature relevant to customer churn. It begins by discussing the definitions and types of churn, with a particular

focus on their relevance in the B2B SaaS industry. Then, the chapter explores the different reasons why customers might leave, based on past research. It also looks at different ways to understand and predict churn, like qualitative methods, traditional statistical models, and machine learning techniques. Finally, this chapter presents the theoretical framework and exploratory hypotheses that guide this study and discusses the thesis's contribution by identifying research gaps and positioning the study within the existing body of knowledge.

Chapter 3 explains the methodology used in this study. It describes the case study design, introduces the case company, and explains the data sources and variables used for the analysis. This chapter also covers the data preprocessing steps, feature engineering techniques, and the process for handling imbalanced data. The selected churn prediction models are presented, along with a discussion of ethical considerations related to their use. Finally, the metrics for evaluating model performance and the methods for model interpretation are outlined.

Chapter 4 presents the findings from the empirical analysis. This includes the performance comparison of the different predictive models implemented in this study, highlighting which models were most effective in predicting customer churn with the case company's data. This chapter will also present the key churn drivers identified through model interpretation techniques.

Chapter 5 explains the results presented in Chapter 4. The findings will be discussed by comparing them to what was found in the literature review in Chapter 2 and the exploratory hypotheses set in section 2.6.2. This chapter also considers the practical implications of the findings for the case company, acknowledge the limitations of this study, and give suggestions for future research. Finally, this chapter sums up the main findings and key points of the whole thesis. It will give final thoughts on the research questions and what the study contributed overall.

2 Literature review

Customer churn refers to an event where a customer stops doing business with a company. This can occur for various reasons, including dissatisfaction with the service, better offers from competitors, or changes in the customer's circumstances. The definition of churn is highly dependent on context and varies across industries and business models. There is no universal definition of churn that works for all industries. Instead, it must be tailored to the specific objectives and customer dynamics of a business (Mutanen, 2006). For example, Rautio (2019) explained that in the Software-as-a-Service (SaaS) industry, churn is often identified when customers cancel their subscriptions or significantly reduce their platform usage. In contrast, Mutanen (2006) suggested that in retail banking, churn might be defined by the closure of a customer's current account, which marks a substantial change in their relationship with the bank. Both studies agree that businesses need to define churn in a way that fits their specific needs. This means looking at the type of customers they have, the industry they are in, and what their business wants to achieve. By understanding these factors, businesses can develop precise churn metrics that reflect their goals and further create strategies to retain their customers. (Mutanen, 2006; Rautio 2019.)

2.1 Types of churn: definitions and relevance in SaaS

Customer churn generally means a customer stops their relationship with a company. To better understand why churn happens and what it means for businesses, churn can be divided into different types (Mutanen, 2006; Rautio, 2019). The exact meaning of a "churn event" can change depending on the industry and business type. For example, in the SaaS industry, churn often means a customer cancels their subscription. In banking, it might mean closing an account (Rautio, 2019; Mutanen, 2006). Even with these differences, the reasons behind churn often help to classify it into distinct types. Understanding these types is important for creating good strategies to keep customers, especially for B2B SaaS companies.

A common way to categorize churn in research is based on who starts the churn process and what the customer's intention is. Many studies separate voluntary churn, where the customer actively chooses to stop using the service, from involuntary churn. Involuntary churn happens because of reasons the customer cannot directly control, or it is started by the company itself (Dahlén & Maurizon, 2023; Peddarapu, 2022; Khare & Arora, 2024).

Lazarov and Capota (2007) provide a more detailed classification, dividing churn for service providers into three main types:

- Active or Deliberate Churn: This is when customers intentionally decide to end their contract and often switch to another provider. Reasons can include being unhappy with the service quality, price, or finding a better offer from a competitor (Lazarov & Capota, 2007). Rautio (2019) calls this “deliberate churn”.
- Rotational or Incidental Churn: In this type, customers end their contract without planning to switch to a competitor. This often happens because their circumstances have changed. For example, they might have financial problems. In a B2B SaaS situation, a customer company might go bankrupt, be bought by another company, or have major internal changes that make the service no longer needed (Lazarov & Capota, 2007; Dahlén & Maurizon, 2023). Rautio (2019) refers to this as “incidental churn”.
- Passive or Non-Voluntary Churn: Here, the company itself decides to end the contract. This could be due to the customer not paying or breaking the terms of service (Lazarov & Capota, 2007). Dahlén & Maurizon (2023) also identify this type of “involuntary churn” which can be caused by external factors or by the company.

It is useful to note that Lazarov and Capota (2007) see both active/deliberate and rotational/incidental churn as types of 'voluntary churn' from the customer's viewpoint. This is because the customer, or their situation, leads to stopping the service, even if they do not switch to a competitor. The main difference is whether dissatisfaction or competition drives the churn, or if it is due to external circumstances.

2.1.1 Focus on voluntary churn in prediction

Despite the different ways of categorizing churn, research shows a strong agreement on which type is most important for prediction and retention efforts. Voluntary or deliberate churn is consistently highlighted as the primary target for SaaS businesses (Dahlén & Maurizon, 2023; Lazarov & Capota, 2007; Peddarapu, 2022; Khare & Arora, 2024; Rautio, 2019). The main reason for this focus is that companies can often influence the factors leading to voluntary churn. These factors can include service quality, pricing, product features, or overall customer experience. By identifying customers at risk of voluntary churn, companies can take proactive steps to keep them.

In contrast, involuntary or passive churn, such as churn due to a customer's bankruptcy or a company's own decision to end a contract, is generally seen as harder to predict or prevent with typical retention actions. For this reason, involuntary churn is often excluded from the data used to build predictive churn models (Dahlén & Maurizon, 2023). This study also adopts this approach by focusing on predicting voluntary forms of churn where the company can potentially intervene.

2.2 Additional dimensions of churn

Besides the main types of churn, like voluntary and involuntary churn, research also points to other important ways to understand churn. These give a greater picture, especially for SaaS businesses. These extra ways help to group churn based on things like the customer's contract, how much they use the service, when churn happens in the customer's journey, and how much value the customer brings. Knowing these details is important for making good retention strategies. The next sections and tables show these other main ways to look at churn.

2.2.1 Churn based on contract status and usage

One important way to group churn is by looking at the customer's contract and how much they use the service. This is especially useful for ongoing services like SaaS. Lazarov and Capota (2007) explain that even if a customer does not fully end their contract, there can be smaller signs that they are becoming less engaged. These can be early warnings of churn. Table 1 shows churn types based on contract status and usage.

Table 1: Churn types based on contract status and usage

Based on contract status and usage				
Category	**Description**	**Key aspects and examples**	**Relevance for SaaS churn prediction**	**Key sources**
Total Churn	Official contract termination; customer completely stops using the service.	Customer formally cancels subscription or ends relationship.	The ultimate outcome most churn models aim to predict.	Rautio, 2019
Hidden Churn	Customer not actively using the service despite an active contract.	Low login frequency, minimal feature usage.	Important early warning signal in SaaS. Indicates disengagement and high risk of future total churn. Trackable via usage data.	Lazarov & Capota, 2007

Partial Churn	Service usage significantly decreases, possibly in favor of a competitor.	Customer reduces number of licenses, stops using key features, shifts some activities elsewhere.	Can precede total churn and already cause revenue loss. Also an early indicator for intervention.	Lazarov & Capota, 2007

Hidden churn occurs when a customer is still under contract but is not actively utilizing the service, while partial churn refers to a significant drop in service usage. Both are critical indicators in the SaaS sector. These patterns are often linked to low customer engagement and a increased risk of eventual churn, and they can frequently be tracked through product usage data (Lazarov & Capota, 2007).

2.2.2 Churn based on timing in the customer lifecycle

The timing of customer churn within their lifecycle provides an essential aspect for grasping its root causes and for effectively scheduling retention strategies. Dahlén and Maurizon (2023) present a temporal classification, segmenting churn into short-term, mid-term, and long-term categories. This viewpoint is particularly significant for B2B SaaS companies, given the differences in customer onboarding durations and the time needed for a customer to realize the complete value of a product. Table 2 outlines these temporally defined churn types.

Table 2: Churn types based on timing in the customer lifecycle

Based on timing in customer lifecycle				
Category	**Description**	**Key aspects and examples**	**Relevance for SaaS churn prediction**	**Key sources**
Short-Term Churn	Occurs early in the customer relationship (e.g., first few months).	Issues with onboarding, mismatched expectations, poor initial product fit.	Highlights problems in early customer experience or sales process. Retention efforts may focus on better onboarding/support.	Dahlén & Maurizon, 2023
Mid-Term Churn	Occurs after initial period but before long-term establishment (e.g., 2-12 months).	Customer has experienced core value but may find issues, or needs change.	Critical period for demonstrating ongoing value. Prediction allows for timely intervention before customer is fully disengaged.	Dahlén & Maurizon, 2023

Long-Term Churn	Occurs after customer has been with the company for an extended period.	Evolving needs no longer met, new competitive offerings, accumulated dissatisfaction.	May require different retention strategies, focusing on loyalty, upgrades, or addressing long-standing issues.	Dahlén & Maurizon, 2023

For instance, short-term churn may suggest problems with the onboarding process or unrealistic expectations established during sales. In contrast, mid-term and long-term churn might reveal evolving customer needs, growing dissatisfaction over time, or changes in the competitive landscape (Dahlén & Maurizon, 2023). Identifying the moments when churn is most likely to occur enables companies to plan strategically and take action accordingly.

2.2.3 Other considerations in churn analysis

Finally, in addition to the direct classifications of churn events, there are broader factors, like the financial value of a customer, that influence how companies prioritize their churn management strategies. Table 3 highlights one such key consideration.

Table 3: Other considerations in churn analysis

Other considerations				
Category	**Description**	**Key aspects and examples**	**Relevance for SaaS churn prediction**	**Key sources**
Customer Lifetime Value (CLV) in Churn	Prioritizing churn prediction/retention for high-value customers.	Not all churning customers have equal financial impact.	Important for resource allocation; focus retention efforts where ROI is highest. Often combined with churn probability.	Lazarov & Capota, 2007

Prioritizing retention efforts based on Customer Lifetime Value (CLV) is a common approach. Not all churning customers have an equal financial impact, and focusing resources on retaining high-value customers can maximize the return on investment for retention initiatives (Lazarov & Capota, 2007).

2.3 Factors influencing customer churn in SaaS industry

Customer churn in the SaaS industry is influenced by many factors, which previous studies often group into subject specific categories. These include product-related factors, customer experience, pricing and contract related reasons, usage patterns, and customer characteristics (Rautio, 2019). Ahn et al., (2006), for example, presented a model with five broad categories of

factors influencing churn: customer dissatisfaction, customer related characteristics, customer status, service usage, and switching costs. Similarly, Dass & Jain (2011) identified both direct factors influencing churn, such as perceived value and service quality, and indirect or potential factors, such as customer loyalty, emotional attachment, service usage, customer demographics, product fit, customer lifetime value and competition. Table 4 summarizes the commonly mentioned factors influencing customer churn in SaaS businesses. It also provides a definition of each factor, as well as the author(s) that have discussed them in the literature.

Table 4: Factors influencing customer churn in SaaS businesses

Factor	Definition	Author(s)
Core Service Failures	Critical incidents caused by provider mistakes or technical problems that disrupt the primary service	Keaveney (1995); Wangenheim et al. (2017)
Service Quality	Overall quality of the service provided (e.g. performance, reliability, uptime)	Szucs & Kiss (2013); Apostolov, (2020); Dass & Jain (2011)
Support Experience	Failures in customer service interactions. (e.g. unhelpful support or poor issue resolution during contact with support)	Keaveney (1995); Kamalraj & Malathi (2013)
Customer Dissatisfaction	The customer's overall dissatisfaction with the product or service, often due to unmet expectations	Ahn et al. (2006); Kamalraj & Malathi (2013); Rautio (2019)
Perceived Value	How valuable the customer perceives the product or service	J. W. Liu et al. (2015); Dass & Jain (2011)
Pricing	The suitability or fairness of the service's pricing for the customer	Keaveney (1995)
Attraction by Competitors	When competitors offer better features or lower prices, customers may be tempted to switch to their services	Szucs & Kiss (2013); Keaveney (1995)
Service Usage	The level of product use, can be measured by frequency of use, duration of use, or number of key actions performed	Buckinx & Van den Poel (2005); Kamalraj & Malathi (2013); Dass & Jain (2011)

Switching Cost	The cost of switching to another service, including losses of integration or benefits by leaving	Heide & Weiss (1995); Kamalraj & Malathi (2013); Dass & Jain (2011)
Customer Loyalty	The customer's commitment to continue using the service, often built through positive experiences and trust over time.	C.-T. Liu et al. (2011); Dass & Jain (2011)
Emotional Attachment	A customer's affective commitment or positive emotional feelings toward the provider	Roos & Friman (2008)
Relationship Tenure	The length of the customer's relationship or subscription with the provider (longer tenure often correlates with higher retention)	Bolton et al. (2004)
Customer Demographics	Characteristics of the customer (e.g. size, sector, age, location)	Dass & Jain (2011); Kamalraj & Malathi (2013)
Technology Orientation	The degree to which the customer is tech-savvy or oriented towards new technology, which can affect adoption and switching behavior	Dass & Jain (2011)
Customer Lifetime Value	The total value the customer represents over the life of the relationship, higher CLV customers tend to stay longer	Rautio (2019)

The variety of factors in Table 4 shows that churn is rarely caused by just one issue. Instead, it usually happens because of the combined effect of the customer's expectations, their experiences, how they see the value of the service, and the pressure from competitors. To manage churn successfully, companies need to take a broad approach that considers all these aspects together. It is also important to recognize that while many factors contribute to churn, their relative importance and interplay can vary significantly depending on the specific context, such as the B2B SaaS environment.

The following chapters will discuss each of these factors in more detail, focusing on their roles, how they affect each other, and what they mean for understanding churn in SaaS businesses.

2.3.1 Product quality and reliability

The quality and reliability of a service or product greatly affect how customers see it and whether they continue to use it. Research clearly shows that product quality and reliability are very important, especially for B2B SaaS companies. For example, Wangenheim et al. (2017) found that service downtimes are often the main reason customers do not renew their SaaS contracts. Keaveney (1995), looking at services more broadly, found that problems with core service features or technical issues were the top reason for customer churn in 44% of cases. This idea is also very important for B2B SaaS. In a B2B SaaS case study, Apostolov (2020) reported that 27% of customers who churned left because the product did not give them the value or performance they expected. Rautio (2019) also showed how important quality is: when quality measures were removed from a churn prediction model, the model's error rate went up by 28.5%.

The needs of B2B customers often make quality even more important. B2B SaaS solutions are usually a key part of a customer's main business activities. This means they need very reliable service, almost with 'zero-downtime' (Apostolov, 2020). Additionally, the complexity of B2B environments, often involving integration with other systems and specific customization needs, means that quality issues in these areas can be particularly disruptive and may contribute significantly to churn.

While most research agrees that quality is essential, there are some details to consider. For instance, Apostolov (2020) found that platform stability was very important, and users were frustrated with technical issues. However, 'reporting dissatisfaction' was not itself one of the top three reasons for churn in their user survey. This might mean that how customers experience quality problems, how much these problems affect their main tasks, and how the company talks about these issues can also matter.

Poor product quality often interacts with other churn drivers. For instance, frequent technical issues can strain customer support resources and negatively affect service usage. Ultimately, a product that fails to deliver on its promises can diminish the customer's perceived value, potentially leading to increased price sensitivity and consideration of alternatives.

Even though product quality is known to be important, the reviewed literature suggests that some areas need more research in B2B SaaS churn studies. For example, more systematic ways to measure how specific parts of quality, like how well the product integrates or how easily it can adapt to new business needs, affect churn rates could be explored. Also, more long-term studies in the SaaS field could help to understand how service quality might change over a long customer relationship, for instance, if performance gets worse over time. So, a quality product that consistently meets customer needs is key to making customers more satisfied and less likely to switch to a competitor.

2.3.2 Customer service and support experiences

The quality of customer service and support also significantly influences customer churn. Keaveney (1995) found that poorly managed interactions between customers and employees often drive customers away. This observation is highly relevant in the SaaS context, where studies indicate that weak support or poor onboarding experiences contribute to higher churn rates. For instance, Wangenheim et al. (2017) highlighted service failures, which can encompass support deficiencies, as a common reason for churn. Apostolov's (2020) B2B SaaS case study showed that 14% of churned customers were dissatisfied with the customer service they received, and Rautio (2019) identified low-quality support as one of the top three main drivers of churn.

Conversely, effective support can play a crucial role in customer retention. Wangenheim et al. (2017) found that positive customer support experiences increase the likelihood of retention. They noted that even small improvements in service quality or responsiveness can positively impact contract renewal rates. When customers feel they receive good service and perceive the provider as responsive to problems, they are more inclined to remain loyal. Unresolved complaints or consistently poor service experiences, however, tend to reduce trust and increase the probability of churn.

While the general impact of service quality on churn is clear, specific industry contexts can introduce variations. For example, Dass and Jain (2011) noted that in the telecom sector, where support processes might be more standardized, customer support was a less critical churn driver. However, for most B2B SaaS companies, providing tailored and effective support is generally considered essential for maintaining customer relationships.

2.3.3 Pricing and perceived value

Economic factors, especially pricing and what customers feel they get for their money, are often mentioned in research as important reasons for customer churn. If customers think the price of a service is too high for what it offers, they are more likely to think about leaving (Rautio, 2019). Kamalraj and Malathi (2013) say that pricing must match what customers expect. They also found that pricing is one of the biggest reasons for churn. In B2B SaaS situations, Wangenheim et al. (2017) also found that price and cost issues are key reasons for churn. Keaveney (1995) noted that pricing problems caused churn in 30% of B2C cases, making it the third most common reason. However, how much price directly affects churn can change. For example, Apostolov (2020) reported that in their B2B SaaS study, only a small number (3%) of customers who churned left due to "not enough resources", which could be related to price. This suggests that other reasons might have been more important in that particular study.

Price vs. Value in B2B SaaS churn

Price is often named as a key reason for customer churn, but how much it directly matters can change. In B2C services, Keaveney (1995) found pricing issues caused 30% of churn. However, a B2B SaaS study by Apostolov (2020) showed only 3% of customers left because of "not enough resources," a reason that can link to price.

This difference suggests that for B2B SaaS companies, things like the total value customers feel they get, product quality, or service may be more important than price alone. It could also mean that B2B customers see price differently when they think about the business benefits they receive.

Very closely connected to pricing is the idea of perceived value. This means how much benefit customers believe they get from a service compared to its cost. Liu et al. (2015) point out that perceived value is a key factor. They found that if customers feel the perceived value of a SaaS service is low for its price, they are more likely to leave. This link between price and value is very important. A high price might be okay if customers also feel they get a lot of value. But if customers think the price is too high for the value they get, they can quickly become unhappy.

The B2B situation can create special conditions for pricing and how value is seen. Pricing plans in B2B SaaS can be complicated. They can range from fees for each user to different levels of company licenses, and often include discussions about the price. This can affect both the actual price paid and how fair customers think the price is, and how much value they feel they get. If pricing can change based on how much the service is used or the size of the company, and if these plans are clear and easy to understand, this can also affect how B2B customers see the value.

Competition in the market can make the effect of price and perceived value even stronger. If a competitor offers better prices for similar features, or more features for the same price, customers are more likely to think about switching (Rautio, 2019). Good offers from competitors are therefore often given as a reason for changing providers (Keaveney, 1995).

The connection between price, perceived value, and other things like product quality is very important. As mentioned in Section 2.3.1, if the product quality is poor, customers will likely feel they are getting less value from the service. This can make them more concerned about its price. On the other hand, sometimes good pricing or discounts might make up for small quality problems for a short time, but this is usually not a good long-term plan. For SaaS providers, this means it is important to offer clear value for money. They need to make sure pricing matches the benefits customers get, and possibly change pricing plans based on how much customers use the service and what

they need. Customers who feel they are paying too much for what they get will probably look for other options.

Even though most research agrees that price and perceived value are generally important, some specific topics could be studied more in the B2B SaaS field. For example, there is not much research on how newer pricing plans, for example usage based pricing, affect customer churn in B2B SaaS companies.

2.3.4 Usage and engagement

How customers use a SaaS product and how engaged they are with it are known to be strong signs of possible churn. If customers use a service very little, it often means they are not getting enough value from it or find it hard to use. This makes them more likely to leave. (Ahn et al., 2006.) The importance of usage data in predicting churn in B2B SaaS is very clear in research. For example, Rautio (2019) found that how much customers used the platform was the most important factor for predicting churn in their study. When this usage data was removed from their prediction model, the model made 29.8% more errors. Similarly, Wangenheim et al. (2017) pointed out that how useful and easy to use customers find a product strongly predict if they will renew their contract. Apostolov's (2020) case study also supports this. It found that the second most common reason for churn (for 17% of customers who left) was that the customer company did not have enough time or enough employees actively using the software. This shows that just having the service is not enough, it needs to be actively used by many people in the customer's organization.

While low usage often points to churn, some studies show a more complex picture. Ahn et al. (2006) suggested looking at how often, for how long, and what key things customers do in the software. Interestingly, their study on telecom services also found that the “billed amount” sometimes linked to higher churn among very active users. This might mean that very high usage could also lead to churn in some cases, perhaps by showing problems with the service or causing other kinds of unhappiness. Dass and Jain (2011), also in a telecom study, said usage could be a churn factor but not one of the top three, showing its importance can change. Keaveney (1995) also put usage in a less important "other" category, but this could be an older view or from a different type of service than modern SaaS.

Is high product usage always a good sign?

Low product usage is often a clear warning sign for churn in SaaS companies (Rautio, 2019). However, the connection between usage and churn is not always straightforward. For example, some studies in the telecom sector found that usage was not a top factor for churn (Dass & Jain, 2011). Separately, Keaveney's (1995) broader study of service industries also placed usage in a less critical category. Even more surprising, Ahn et al. (2006) discovered that very high usage sometimes led to more churn for telecom customers. This was often because high usage was linked to high, usage-based bills. In such cases, a competitor offering a flat-rate, unlimited plan could be very attractive to these high-usage customers.

This variability highlights that the impact of usage depends on industry and pricing. For B2B SaaS, understanding how a product is used and its value relative to cost is key, not just usage volume. This study's model interpretation methods can help clarify these complex links between usage, value, and churn.

Product usage is often connected to other reasons for churn. As said before, low usage can mean customers do not see the product as useful. This might be because of poor product quality (Section 2.3.1) or not enough support (Section 2.3.2), leading to unhappiness and finally churn. On the other hand, as Ahn et al.'s (2006) finding suggests, very high usage might show problems with how well the product can handle many users, or issues with its quality or support, leading to churn in a different way.

All in all, it is very important for SaaS providers to track how much their products are used. This helps find customers who might not see the service as very important and could be at a higher risk of leaving (Rautio, 2019). However, more research could give us a better understanding in the B2B SaaS area. For example, a more detailed look at how the use of specific features links to churn for different types of B2B customers could give useful information for improving products and keeping customers.

2.3.5 Switching costs and contractual barriers

How easy or difficult it is for customers to switch to another service or provider also affects customer retention. High switching costs usually make customers less likely to leave because changing providers becomes less appealing. These costs are not just direct money penalties or fees for ending a contract early. They also include the large amount of time and effort needed to move data, learn a new system, train employees, and build new relationships with a different provider (Dahlén & Mauritzon, 2023). For B2B SaaS, Blut et al. (2015) pointed out that switching costs also mean researching and

checking other options, handling technical connections, and worrying if a new solution will actually be better.

Switching costs are often higher and more noticeable in B2B situations. B2B SaaS products are usually deeply connected to a customer's daily work through complex integrations and specific setups. These things, along with possible long-term contracts or "lock-in" terms, tend to make switching more costly for businesses than for individual users. The study by Blut et al. (2015) found that losing established relationships and trust and the effort and time needed to learn new ways of working were strong reasons stopping customers from switching. Similarly, things like loyalty programs or product features that make it hard to leave can also prevent customers from changing providers (Ahn et al., 2006; Dass & Jain, 2011).

If switching is hard or expensive, it naturally reduces churn because customers do not want to go through the extra work and cost. However, switching costs do not always work in a simple way. Ahn et al. (2006) noted that very high switching costs can make customers unhappy if they feel "trapped" with a provider, especially if they are not satisfied with the service or its value. This shows that switching costs can have two sides: they can keep customers, but they can also cause bad feelings if the customer does not see much benefit in staying. Adding to this complexity, Ahn et al. (2006) found something unexpected: even though loyalty points usually lowered churn, members of a certain "membership card" program churned more often than expected. This might be due to the type of customers who joined that program or other unobserved reasons. The study by Blut et al. (2015) also suggests that different kinds of switching costs can affect how customer satisfaction relates to future purchases in different ways, showing it is a complex issue.

Besides the costs of actually switching, contract terms like how long the contract lasts and how it is renewed also affect churn. Bolton et al. (2004) argued that longer contracts might make customers more likely to stay, partly because clients can become more connected and "loyal" over time as the relationship grows. So, while making switching easier could increase churn, long-term contracts or services that are deeply integrated often create bigger barriers that stop customers from leaving.

Even though we know switching costs are important, more research could help us understand them better in the B2B SaaS field. For example, it would be useful to have better ways to measure how much complex integrations, different kinds of vendor lock-in, and specific technical switching costs affect churn.

2.3.6 Competition and alternative offerings

The presence of competitors in the market also affects customer behavior and churn. Customers might switch if a competitor's offer seems more attractive due to better features, service, or lower costs (Keaveney, 1995). Keaveney

(1995) called this the “attractiveness of competitors”, meaning how good competing services look compared to what the customer currently uses. In a competitive market like B2B SaaS, customers might not leave because their current provider is bad. Instead, they might leave because a competitor offers something better, like new technology or lower prices.

However, studies show different views on how much direct competition leads to customer churn. For example, Apostolov's (2020) B2B SaaS study reported that only about 2% of customers who churned said they switched directly to a competitor, suggesting direct competition was not a major stated factor in that specific case. Some earlier general service perspectives also viewed competition as less central (Keaveney, 1995). This contrasts with other findings. For instance, Dass and Jain (2011) found in the telecom sector that “attraction by competitors” caused churn in about 10% of cases. Similarly, Ahmed and Linen (2017) identified more enhanced or affordable services from other providers as a major cause of churn, highlighting the role of competitors.

Understanding the impact of competition

Research shows different findings on how much direct competition leads to customer churn. For example, in a B2B SaaS study, Apostolov (2020) found only 2% of customers left mainly due to a competitor. However, Dass and Jain (2011) reported that competition was the reason for about 10% of churn in the telecom industry, and Ahmed and Linen (2017) also pointed to competitor offers as a major churn driver.

These differences could be because of the industry type (SaaS vs. telecom), the specific market conditions, or how the reasons for churn were studied. This shows we need to look deeper than just what customers state as their primary reason.

This study uses model interpretation tools. These tools can help find out if competition has a hidden, larger role or how it works together with other factors in a B2B SaaS company.

In the B2B SaaS field, the effect of competition can be complex. It is not just about comparing features or prices directly. Things like business partnerships and existing relationships in the industry can also influence customer choices. Because there are other options available, service providers must keep improving and meeting market expectations to keep their customers.

In addition, competition can make other churn drivers seem more important. For example, if there are many competitors, customers might become more sensitive to the price of their current service or more critical if the

product quality is not good. Competition can also make switching to another provider seem easier if a competitor offers help or good deals for switching (related to Section 2.3.5 on switching costs). Therefore, even if switching directly to a competitor is not always the most common reason given for churn, it is important for companies to offer unique features and good value. If a company falls behind competitors in quality or price, customers are more likely to look at other options. More research on how specific ways of competing, like offering different features or making alliances with partners, affect churn in B2B SaaS could be useful for businesses.

2.3.7 Dissatisfaction, loyalty, and relationship tenure

Customer dissatisfaction is a factor mentioned in almost all studies looking at reasons for customer churn. According to Ahn et al. (2006), when customers are not satisfied with a product or service, they are very likely to switch to another provider. This is true unless high switching costs make it difficult to change. Even then, if a customer's main needs and expectations are not met, high switching costs and contract terms might only delay, not prevent, churn. Ahn et al. (2006) also found that the number of complaints and unhappiness with unmet service promises directly lead to churn. Similarly, a study by Blut et al. (2015) showed that customer satisfaction is the strongest factor driving customers to buy again from the same company.

On the other hand, strong customer loyalty can reduce the chance of customers switching or churning. Loyalty usually grows over time through repeated good experiences and trust. Liu et al. (2010) describe loyalty as the customer's willingness to keep using the services of the same provider. Loyal customers often forgive small mistakes and are also less interested in offers from competitors. However, it is important to note that loyalty does not always protect against churn. Some satisfied and loyal customers might still switch for reasons like convenience, a very attractive offer from a competitor, or price considerations.

Dass and Jain (2011) found that both loyalty and emotional attachment can act as indirect barriers stopping customers from churning. Roos and Friman (2008) also pointed out that emotional reasons can make it harder for customers to change providers, even when they are not completely satisfied. So, emotional factors, like how a customer feels about the service provider and their overall satisfaction, play an important role.

Dissatisfaction is often connected to other churn drivers already discussed, such as poor product quality (Section 2.3.1), inadequate customer support (Section 2.3.2), or issues with pricing and perceived value (Section 2.3.3). Problems in these areas can directly lead to customer dissatisfaction.

As customers gain more experience with a company, the length of their relationship itself can also increase loyalty. Customers who have been with a company for a long time have often invested more in the relationship and are

usually more hesitant to switch. Bolton et al. (2004) describe the duration of the customer relationship as one part of customer assets. They found that the longer a customer stays with a company, the more likely they are to continue the relationship and bring more value to the business over time. This is particularly relevant in B2B contexts where long contract periods are common, and the financial impact of losing a long-term customer can be significant.

While the roles of satisfaction and loyalty are widely studied, some aspects could be explored further in the B2B SaaS context. For example, the specific role of emotional attachment or brand attachment in B2B SaaS customer relationships and its impact on churn is an area with limited research.

2.3.8 Characteristics and demographics

Research shows that certain things specific to customers can also affect if they are likely to churn. These are not directly about the product or service itself, but they change how customers use and feel about the service. For example, in B2B SaaS, things like company size, the type of industry the company is in, or how good they are with technology can be linked to different churn rates (Rautio, 2019). Dass and Jain (2011) noted that in service businesses, both demographics and how a customer feels about technology could affect churn. For instance, customers who are good with technology might try new tools more easily. They might also churn more quickly if they find a newer, better solution. On the other hand, customers with less technical skill might find it harder to start using a new service and could then be more at risk of churning. Similarly, Ahn et al. (2006), in a study about telecom, found that things like customer grade and status were direct signs of churn risk, showing that a customer's overall profile can be important.

It is important to know which kinds of characteristics are best at predicting churn. Traditional information like age or gender, which are often used in B2C churn studies, have sometimes been found to be less good at predicting churn compared to data about how customers behave, like service usage (Khan et al. 2010). In B2B SaaS, information about the company and signs of how ready they are for technology or if they have enough resources are often more useful. For example, small startup companies might churn more often because their budgets are not stable or they do not have enough staff. Bigger companies might stay longer because their needs are more stable.

Another important idea often discussed with customer characteristics is customer lifetime value (CLV). CLV measures the total value a customer is expected to bring to a company during their whole time as a customer. CLV is more an result of keeping customers for a long time than a direct reason for churn, but it is often connected to churn risk. Customers with a high CLV often use the service more and might get more attention from the company. This can, in turn, lower their chance of churning (Rautio, 2019).

Understanding the customer's profile helps companies to better predict churn and plan how to retain customers. These characteristics might not usually be the main reasons for churn by themselves, but they can change how other factors affect the customer's experience. Knowing these differences helps companies to create better plans, like giving more support when new customers from certain groups start using the service if they might be at higher risk. However, more detailed research on different customer groups in B2B SaaS could give even more useful information for managing churn. This could include looking at things like where the customer is in their own business journey or using more complete “account health scores”.

2.3.9 Summary and comparative overview of churn drivers

The sections 2.3.1 to 2.3.8 looked at many different single factors that can affect customer churn in SaaS businesses. This included product quality, customer service, pricing, and how customers use the product. This review shows that while research often names many of the same factors, how much each factor matters, how they affect each other, and which ones are most important can be different. It depends on the specific SaaS product and its customers. It seems that things like how actively customers use the product, the quality of the service, and the special ways B2B customer relationships work are often very important.

To show more clearly how the importance of different churn drivers can appear in actual B2B SaaS research, Table 5 gives a summary of the main churn drivers or predictive factors found in some selected studies. This comparison shows common ideas but also differences in what studies found, depending on the type of study and how they found the drivers.

Table 5: Summary of main churn drivers in previous studies

Study	**Context of study**	**Key churn drivers / Predictive factors found**	**How drivers were found & key observation**
Rautio (2019)	B2B SaaS (Finland)	1. How much platform was used (time spent, active users) 2. Service quality information (support ratings, NPS) 3. Business information (customer ad spending on platform)	Used Random Forest model to see feature importance (error increase if feature removed). Usage data was most important (29.8% error increase).

Apostolov (2020)	B2B Data Management SaaS	1. Company did not keep promises (27%) 2. Customer had too few resources (time/staff) (17%) 3. Unhappy with customer service (14%)	Based on a questionnaire to customers who churned. Showed a gap between sales promises and what happened, and that customers need to put in effort.
Dahlén & Mauritzon (2023)	B2B Enterprise SaaS	1.Customer tenure 2. Number of projects 3. Total activity in expense reports	Used ML models (SHAP values) to find important features. These factors were important for both short & long-term predictions. Long tenure could mean higher risk.
Wangenheim et al. (2017)	IT-based Service Contracts (B2B focus)	1. Past usage behavior (how much, how widely, how long) 2. How customers see the technology (Perceived usefulness, Perceived ease of use) 3. How usage & perceptions connect	Used statistical modeling for renewal decisions. How useful customers see the service becomes more important the longer/more they use it.
Khare & Arora (2024)	SaaS Products (B2B focus)	1. Time Spent on Platform (40% importance) 2. How often users log in (32% importance)	Used ML model to see feature importance. Showed strong focus on how much users directly engage with the product.

As Table 5 shows, some drivers like product usage and service quality are mentioned often. However, different studies might focus on slightly different things or use different ways to find the key drivers. This variety shows that churn is a complex topic. It suggests that even though there are general ideas that apply to many situations, companies often need to do their own analysis to find the most important churn signs in their own business. This thesis does that kind of company-specific analysis. Understanding these factors is the first step. The next step is to look at the ways and methods used to predict churn and, in the end, reduce it. These will be discussed in the following section.

2.4 Approaches to understanding and predicting customer churn

Understanding why customers churn and being able to predict when they are likely to leave are both important for improving customer retention. Previous research has looked at churn from two different but connected perspectives. Some studies have focused on qualitative research to explore the reasons behind churn, while others have used quantitative methods to estimate the risk of churn (Rautio, 2019). In a B2B SaaS environment, churn is especially critical because recurring revenue and long-term customer relationships are central to business success. This section will review these different methodological approaches.

To provide an initial overview of the main approaches discussed in this section, along with their typical strengths, limitations, and some of the key research gaps identified in the literature regarding their application to customer churn, Table 6 offers a summary. The following sections will then delve deeper into each of these categories: qualitative methods, traditional statistical modeling, and modern machine learning techniques, highlighting their roles and challenges in churn analysis.

Table 6: Different approaches used in churn prediction

Method	Primary goal in churn analysis	Typical strengths in churn context	Limitations in churn context	Gaps highlighted in literature
Qualitative approaches	Understanding the 'why' of churn, exploring reasons & context	Rich, deep insights into customer experiences; identifying complex/unexpected reasons; hypothesis generation. (Keaveney, 1995)	Small samples, generalizability issues; difficult to quantify for large-scale prediction; potential for bias in self-reported data. (Rautio, 2019; Keaveney, 1995)	1. How to best integrate rich qualitative insights with quantitative predictive models. 2. Satisfied customers may still churn (Keaveney, 1995)

Traditional statistical models (e.g., Logistic Regression)	Predicting churn probability, identifying general risk factors	Relatively simple to implement & interpret (coefficients); provides churn probabilities for targeting. (Mutanen, 2006; Günther et al. 2014)	Assumes linear relationships; struggles with complex interactions & nonlinearities; multicollinearity can hinder interpretation; static models may require frequent updates. (Mutanen, 2006; Sabbeh, 2018)	1. Predictive accuracy vs. depth of causal understanding (Mutanen, 2006). 2. Performance often lower than ML with complex data. 3. Handling class imbalance effectively.
Machine learning (ML) approaches	Improving predictive accuracy with complex data	Better at handling large, complex, nonlinear data; often higher predictive accuracy; some models offer feature importance. (Rautio, 2019; Sabbeh, 2018)	Can be "black boxes" (low interpretability); require more data & feature engineering; potential for overfitting; computational resources & expertise needed. (Rautio, 2019)	1. Predictive power vs. interpretability (the "black box" problem). 2. Performance of advanced models (e.g., deep learning) highly dependent on data volume (Rautio, 2019). 3. Need for robust methods for model explanation (SHAP).

2.4.1 Qualitative approach: Why do customers churn

One way to explore customer churn is through qualitative research. Methods like interviews, focus groups, and exit surveys allow companies to ask customers directly why they left or were thinking about leaving. These techniques can reveal deeper issues and the underlying 'why' of churn, which might not be obvious from numbers alone.

A study by Keaveney (1995) shows how valuable such qualitative insights can be. Keaveney's research used the Critical Incident Technique (CIT) to understand the behavior of service customers who had switched providers, collecting detailed accounts from over 500 people. As discussed in the previous chapter, this analysis identified common factors influencing customer churn, including service failures, poor customer experiences, and pricing problems. For instance, customers mentioned specific incidents like billing mistakes, unfriendly staff, or unfair prices as reasons for leaving (Keaveney, 1995). This

highlights that churn can be caused by a variety of issues, some going beyond the product itself, such as an employee's attitude. Keaveney (1995) argued that such a qualitative approach is particularly useful when a topic is not well understood, as it allows for the development of new understanding directly from customer experiences.

Qualitative research is helpful for creating new insights and identifying patterns in customer feedback. For example, if many customers mention slow support, that factor can later be tested as a predictor in a churn model. However, as Rautio (2019) noted, qualitative studies are often based on small samples or individual stories, which may not apply to the entire customer base. This can limit the generalizability of the findings (a limitation also acknowledged by Keaveney, 1995, regarding their sample). It is also difficult to turn detailed feedback from interviews directly into a form that predictive models can use for future churn prediction. Rautio (2019) added that qualitative data is usually unstructured and not suited for large-scale churn scoring. In addition, customers might not always explain their reasons completely or accurately, which could lead to biased or incomplete information (Rautio, 2019). Keaveney (1995) also pointed out that focusing on specific incidents might miss the impact of gradually accumulated negative experiences.

A key insight from qualitative explorations like Keaveney's (1995) is that even customers who report being generally satisfied might still churn due to other reasons such as convenience, attractive competitive offers, or pricing. This presents a 'tension' or a point of contrast with purely quantitative models that might heavily rely on satisfaction metrics as the main predictor of loyalty. Qualitative methods can uncover these more complex decision-making paths.

The insights from qualitative research should be seen as a useful starting point. They highlight important factors and give context that helps in building effective predictive models. As Ahmed and Maheswari (2017) point out, understanding why customers leave is important, but it should ideally be supported by data analysis to actually predict and prevent churn on a larger scale. Therefore, companies often benefit from combining qualitative insights with more quantitative methods. For instance, a previous qualitative study within the case company provided initial understanding of potential churn drivers, which helped to inform the selection of variables for the quantitative modeling in this thesis.

2.4.2 Statistical modeling for churn prediction

On the quantitative side, many early churn prediction studies relied on traditional statistical methods, with logistic regression being a particularly common choice (Günther et al., 2014). Logistic regression estimates the probability of a customer churning based on various input variables, such as service

usage, contract length, or support history. In the model, each variable is assigned a coefficient that indicates how it affects churn risk: a positive coefficient suggests an increased likelihood of churn, while a negative one suggests a reduced risk (Stucki, 2019). This approach has been popular due to its relative simplicity and the perceived ease of interpreting its results (Günther et al., 2014). For instance, if a variable like 'months since last login' has a strong positive coefficient, it clearly signals that inactive users are more likely to churn. Mutanen (2006), in a study within the banking sector, found that logistic regression worked well to identify which customers were at the highest risk of leaving, using factors like the length of the customer relationship or the number of bank products. Mutanen (2006) also highlighted that a key strength of logistic regression is its ability to output a churn probability for each customer, which can be valuable for businesses when targeting retention efforts with limited resources.

Although logistic regression can offer insights and serve as a good starting point, it also has notable downsides and limitations, especially when applied to complex churn data. One main limitation is its assumption of a linear relationship between the predictors and the log-odds of churn, which is not always true for complex customer behaviors. Customer churn is often influenced by intricate interactions among multiple factors, and capturing these interactions effectively with a standard logistic regression model can be difficult. As datasets grow larger and more feature-rich, research has indicated that logistic regression may be outperformed in predictive accuracy by more advanced machine learning methods (Sabbeh, 2018).

Furthermore, while the model coefficients can offer some general understanding of churn drivers, interpreting them precisely can be challenging, especially if multicollinearity is present in the data, a difficulty also pointed out by Mutanen (2006) in their study. This creates a tension between the model's predictive output and the depth of causal understanding it can reliably provide about individual factors. Another practical challenge, noted by Mutanen (2006), is that customer profiles and the factors driving churn can change over time. This means that a single, static logistic regression model might not maintain its accuracy for long and would likely require regular updates and recalibration. Additionally, like many predictive models, logistic regression is sensitive to class imbalance, a common issue in churn datasets that needs to be addressed through techniques such as sampling (Mutanen, 2006).

Despite these limitations, logistic regression still serves as a solid baseline model in many churn prediction studies. It provides an excellent starting point for understanding the key factors linked to churn and offers a benchmark against which the performance of more complex machine learning methods can be compared (Stucki, 2019).

2.4.3 Machine learning for improved churn prediction

As companies, especially in the data-rich SaaS industry, collect more detailed customer data, machine learning (ML) techniques have become increasingly common and often preferred for churn prediction. These methods are generally better at handling the complex, non-linear relationships and interactions among many features that are often found in large SaaS datasets. ML algorithms can analyze numerous variables simultaneously and identify patterns that traditional statistical models like logistic regression might miss (Rautio, 2019). This ability to capture more intricate patterns often leads to higher predictive accuracy. Rautio (2019) specifically argued for the need for ML in the B2B SaaS context, pointing to a relative lack of prior research in this specific domain and the strong business need for more accurate and timely churn predictions.

Machine learning has been successfully used for churn prediction in various industries. Multiple case studies focusing on churn prediction highlight that ensemble methods, which combine the predictions of several base models, often achieve better performance. For example, Keramati et al. (2014) used several data mining techniques on telecom customer data, including decision trees, artificial neural networks (ANNs), k-nearest neighbors (k-NN), and support vector machines (SVMs). Their hybrid model, which combined multiple algorithms, achieved high recall and precision (over 95%). Sabbeh (2018) tested ten different algorithms on another telecom churn dataset and found that ensemble methods like Random Forest and AdaBoost reached the highest accuracy (around 96%), clearly outperforming logistic regression, ANNs, and SVMs. Similarly, a review by Dahlén and Mauritzon (2023) of recent churn prediction studies indicated that ensemble methods frequently achieved over 90% accuracy. In their own B2B SaaS churn study, Dahlén and Mauritzon (2023) found that XGBoost (a gradient boosting method) performed best, achieving an AUROC of 0.900, with good accuracy and recall. Khare and Arora (2024) also benchmarked four ML algorithms for a SaaS company and concluded that Random Forest and gradient boosting were favored in most model evaluations, significantly outperforming logistic regression and SVM in their specific case.

While a variety of ML models have been tested, the "best" performing model can depend on the specific dataset and context. For instance, in Rautio's (2019) B2B SaaS study, an SVM model achieved the highest precision, although ensemble methods like Random Forest also showed their utility, particularly in understanding feature importance. This highlights that there isn't always a single "winner" ML algorithm for all churn problems. A challenge also exists with more complex models like deep neural networks. While they might offer high accuracy with very large datasets, Rautio (2019) found they did not perform well with their more limited B2B SaaS dataset,

suggesting that the volume and nature of data are critical considerations for choosing such advanced models.

A significant advantage of some ML models, and an area of increasing focus, is interpretability. While highly complex models can sometimes act as "black boxes", making it hard to understand why a prediction is made, many ML approaches offer tools for interpretation. For example, tree-based ensemble models like Random Forest can provide feature importance scores, indicating which input variables had the most influence on the model's predictions (Rautio, 2019; Khare & Arora, 2024). Rautio (2019) used this to identify key churn driver categories such as platform usage and service quality. More advanced techniques like SHAP (SHapley Additive exPlanations) values are also being used to explain the output of complex models on an individual prediction level (Dahlén & Mauritzon, 2023). Having interpretable results is very important for managers who need to understand the reasons behind potential churn to design effective and targeted retention strategies.

Despite their strengths, applying ML models for churn prediction also comes with challenges and limitations. These include the need for high-quality and comprehensive data, careful feature engineering to create meaningful input variables, and effectively addressing class imbalance, which is a common problem in churn datasets (Mutanen, 2006; Rautio, 2019). Additionally, deploying and maintaining ML models can require specialized expertise and computational resources. Finally, as Rautio (2019) also pointed out, some churn is inevitably driven by external factors not present in the available data (e.g., sudden budget cuts at the customer's company), which sets a fundamental limit on the predictive power of any data-driven method. Future research in this area often focuses on developing more robust feature selection techniques, optimizing individual ML models for specific churn problems, and exploring their application across a wider range of business contexts and datasets (Rautio, 2019).

Table 7 summarizes key studies discussed in this section, highlighting their context, methods, and objectives in tackling customer churn.

Table 7: Overview of key studies on customer churn prediction

Study	Context	Method(s) used	Purpose of study
Keaveney (1995)	Services (Multiple industries, B2C)	Qualitative critical-incident interviews	Identify reasons for customer switching qualitatively, by categorizing common causes of churn in services.

Mutanen (2006)	Retail Banking (B2C)	Statistical model (Logistic Regression)	Predict customer churn and analyze differences between churning and non-churning customers; demonstrate use of logistic regression in churn analysis.
Keramati et al. (2014)	Telecommunications (Mobile, B2C)	Machine learning (Decision Tree, ANN, k-NN, SVM; plus hybrid ensemble)	Compare multiple data mining techniques for churn prediction and propose a hybrid model to improve accuracy in telecom churn detection.
Ahmed & Maheswari (2017)	Telecommunications (B2C, review study)	Literature Review of churn prediction methods (focus on ML & hybrid models)	Analyze and summarize churn prediction techniques; emphasize understanding churn behavior and highlight that hybrid models yield the best predictive performance.
Sabbeh (2018)	Telecommunications (Mobile, B2C)	Machine learning (Comparative study of 10 classifiers: e.g., Random Forest, AdaBoost, Neural Network, SVM, k-NN, Logistic regression)	Compare performance of various machine learning techniques for churn prediction; identify which algorithms most accurately predict churn.
Dahlén & Mauritzon (2023)	B2B SaaS (Software, Enterprise clients)	Machine learning (Logistic Regression, Random Forest, XGBoost, Gradient Boosting, LightGBM)	Develop and evaluate ML models to predict churn in a B2B SaaS product; identify key churn risk factors and select the best-performing model (XGBoost) for accuracy.

Khare & Arora (2024)	SaaS Products (Cloud services, B2B focus)	Machine learning (Supervised learning on customer usage data)	Explore the efficacy of ML models in predicting churn in SaaS; aim to anticipate churn and enable targeted retention strategies, demonstrating ML's value in a SaaS context.

2.5 Evaluating the performance of churn prediction models

After building churn prediction models, it is important to carefully evaluate their performance to see how well they actually work. Simply creating a model is not enough, it needs to be checked with suitable measures to see if it correctly finds customers who might leave, while making few mistakes. (Stucki, 2019.) Evaluating churn models can be difficult because churn data is often imbalanced. This means usually fewer customers churn compared to those who stay. This imbalance can make simple evaluation measures give a misleading picture of performance. Therefore, a more detailed evaluation approach is needed, which considers the company's specific goals for keeping customers. (Fawcett, 2006; Dahlén & Mauritzon, 2023.) This section discusses the key issues and common ways to choose evaluation metrics for churn prediction based on previous studies.

One major challenge comes from using only accuracy. While easy to understand, accuracy alone can be very misleading when churn is rare, which is common in churn datasets. (Fawcett, 2006.) For instance, a model predicting that no one will churn might seem highly accurate if only a small percentage of customers actually leave. However, such a model would be useless in practice because it completely fails to identify any customers at risk. Relying only on accuracy is therefore not enough, especially since failing to identify a churning customer can lead to significant lost revenue. (Fawcett, 2006; Stucki, 2019.)

Stucki (2019) suggested that, to gain more useful insights, especially about the smaller group of churning customers, other metrics like precision and recall should be used. The choice between focusing on precision or recall depends heavily on the business context. A high recall is crucial if the main goal is to identify as many potential churners as possible, even if it means sometimes wrongly flagging loyal customers. Missing a real churner could mean lost income that might have been saved by taking action in time. On the other hand, high precision is more important when the actions taken to keep customers are expensive. In this case, wrongly identifying loyal customers as churners wastes resources and might even annoy those customers. (Stucki, 2019.)

Stucki (2019) added that there is often a trade-off where improving recall might lower precision, and vice versa. Finding the right balance between these two depends on whether it is more costly for the business to miss potential churners or to spend resources on customers who were not actually going to leave. Understanding this trade-off is key to selecting the right decision threshold for the model predictions. (Stucki, 2019.)

Because precision and recall often pull in different directions, metrics that combine them or evaluate performance across all possible thresholds can be very helpful. The F1-score, for example, provides a single number that balances precision and recall. It is particularly useful with imbalanced churn data, as it only gives a high score if both precision and recall are reasonably good. This makes the F1-score a good metric for comparing different models when a balanced performance is needed. (Vafeiadis et al., 2015); Stucki, 2019.)

Another widely used metric that avoids depending on a single threshold is the Area Under the ROC Curve (AUC). A higher AUC indicates the model is generally better at distinguishing between customers who will churn and those who will not. AUC is useful for comparing the overall effectiveness of different models and is less influenced by the class imbalance than accuracy is. (Fawcett, 2006; Dahlén & Mauritzon, 2023; Stucki, 2019.)

While the metrics like F1-score and AUC are useful, understanding the specific types of errors a model makes is also important for practical use. Analyzing the model's errors, specifically looking at how many actual churners were missed (false negatives) versus how many loyal customers were wrongly flagged (false positives) at a certain threshold, gives valuable insight. (Dahlén & Mauritzon, 2023.) Taking a closer look helps businesses figure out if the model is bad at spotting customers who are about to leave or if it is wrongly flagging loyal customers. This understanding can guide decisions about adjusting the model or changing the decision threshold to better align with business strategy. (Stucki, 2019.)

In summary, choosing the right evaluation metrics for churn prediction is important. The choice must match the company's business goals and acknowledge the challenges of working with often imbalanced churn data. Relying on accuracy alone is insufficient. (Fawcett, 2006; Stucki, 2019.) Depending on whether the priority is to catch all potential churners or to be efficient with retention efforts, different metrics become more important. Metrics like the F1-score offer a balance, while AUC provides a threshold-independent view of model performance. Because each metric highlights different aspects of performance, it is strongly recommended to use and report multiple metrics (e.g., accuracy, precision, recall, F1-score, AUC, and analysis of the confusion matrix) to get a complete picture and ensure the chosen model truly supports the company's customer retention strategy. (Stucki, 2019.)

2.6 Theoretical framework and hypotheses

This section outlines the theoretical framework that guides this study and presents the exploratory hypotheses based on the literature review. The framework provides a conceptual model of the key elements involved in comparing predictive methods for customer churn in a B2B SaaS context, while the hypotheses set specific expectations for empirical research.

2.6.1 Conceptual framework

The literature review shows a broad process for understanding and predicting customer churn. To bring these factors together, a general conceptual framework showing the overall research process of this study is presented in Figure 2. This framework shows the links between the case company context, finding churn drivers from literature and data, choosing and using predictive methods, checking model performance, and the final goal of answering the research questions and giving useful recommendations. The case company's situation, with its specific data and business needs, is the base that affects all following stages of this research.

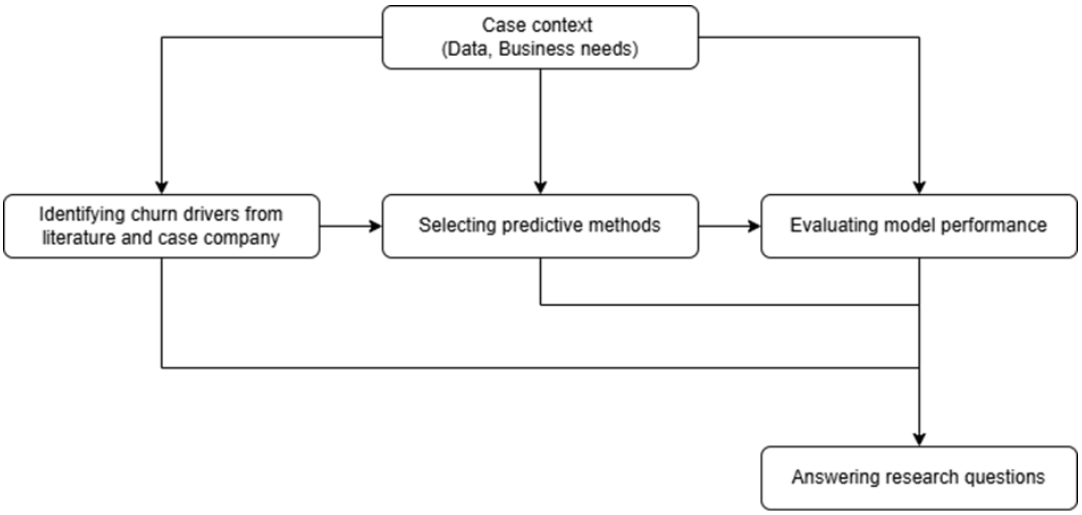

Figure 2: Conceptual framework for comparing predictive methods for customer churn.

While Figure 2 shows the research steps, a more specific conceptual model is needed to understand the connections between different factors and the likelihood of customer churn in the B2B SaaS setting. Based on the detailed review of churn drivers in Section 2.3, and taking ideas from known frameworks in service management and relationship marketing, a conceptual model for B2B SaaS churn drivers is suggested in Figure 3.

This model (Figure 3) suggests that several groups of factors add to the likelihood of churn. Basic theories like the Service-Profit Chain (Heskett et al., 1994) highlight the link between how services are delivered, customer satisfaction, and loyalty, which then helps company profits. In a similar way, Commitment-Trust Theory in relationship marketing (Morgan & Hunt, 1994) points out how important trust and a strong connection are for keeping long-term customer relationships and making customers less likely to leave. While these theories give a general understanding, the model in Figure 3 changes these ideas and adds factors that are specific to B2B SaaS companies.

The model identifies a set of initial or external factors that can affect a customer's journey:

- Product factors: This includes things like product quality, if it works reliably, its features, and ease of use. A good quality product that fits customer needs is expected to improve the customer experience.

- Pricing & contractual factors: This covers how fair the price seems, how clear the contract terms are, and the total financial aspect.

- Service quality: This is about the quality of customer support, the process of onboarding, and ongoing service help.

- Company characteristics & Market position: These are background factors like the customer's company size or industry, and the level of competition in the market, which can directly affect how likely churn is.

These initial factors are thought to affect several middle-stage factors:

- Perceived value: The customer's overall idea of the benefits they get from the service compared to its cost. Product factors, pricing, and even how much they use the product are expected to shape this.

- Customer engagement & usage: How much and how deeply a customer uses the SaaS product. Product factors are key drivers here, and engagement can then affect perceived value.

- Customer satisfaction: How happy the customer is overall with the service. This is largely affected by service quality but also by product experiences and the value they feel they get.

- Relationship strength & trust: The level of trust and strong connection built between the customer and the SaaS provider. This grows from positive experiences and satisfaction.

Finally, these middle-stage factors are expected to directly affect the Likelihood of Churn. Higher perceived value, strong engagement, satisfaction, and a solid relationship are all expected to lower the chance of a B2B SaaS customer churning. On the other hand, problems in these areas are likely to increase churn risk. This model gives a clear way to think about the many sides of churn drivers. These drivers will be looked at through the predictive models and how they are understood in this study.

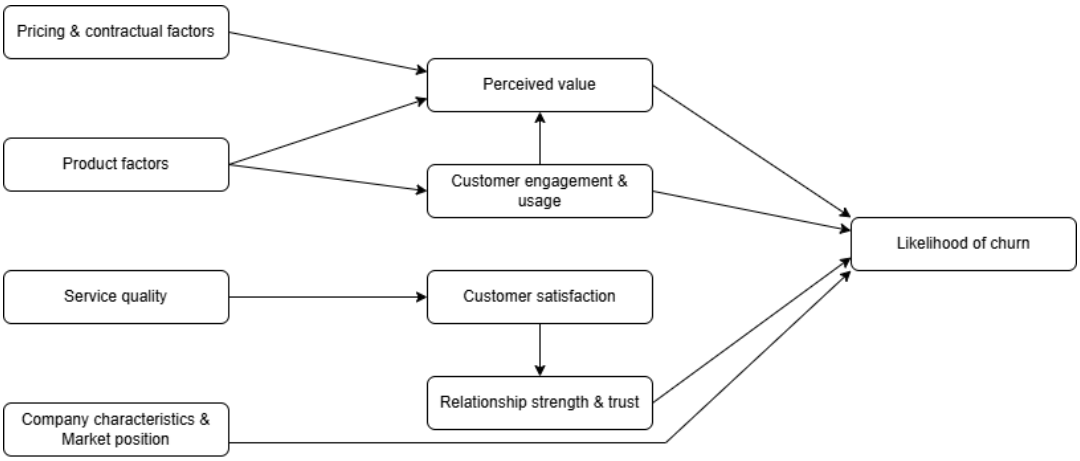

Figure 3: Conceptual model of B2B SaaS churn drivers

2.6.2 Exploratory hypotheses

Based on the literature review and the conceptual framework, the following hypotheses are proposed to guide the empirical part of this study. These are not intended for formal statistical testing in a strict sense, but rather as guiding propositions based on existing research, which this case study will explore in the specific context of the B2B SaaS company.

The first hypothesis relates to the performance of predictive methods and is linked to research question 1. Previous literature often reports that ensemble-based machine learning models perform well in churn prediction (e.g., Sabbeh, 2018; Dahlén & Mauritzon, 2023; Khare & Arora, 2024). Therefore, it is hypothesized that:

- H1: Ensemble-based machine learning models, such as Random Forest and Gradient Boosting, will show higher predictive performance for customer churn compared to logistic regression and potentially other individual machine learning classifiers. Performance will be measured using metrics like F1-score and AUC on the case company's B2B SaaS data.

The second hypothesis concerns the key churn drivers and is linked to research question 2. Based on the conceptual model of B2B SaaS churn drivers (Figure 3) developed from the literature review, it is hypothesized that:

- H2: The main elements highlighted in the conceptual model (Figure 3), specifically Product factors, Pricing & contractual factors, Service quality, Customer engagement & usage, Perceived value, Customer satisfaction, and Relationship strength & trust, will be important indicators of customer churn in the case company's B2B SaaS data.

The third hypothesis is about model interpretability and practical recommendations, linking to research questions 2 and 3. Understanding the "why" behind predictions is important for creating actionable retention strategies (e.g., Dahlén & Mauritzon, 2023). Thus, it is hypothesized that:

- H3: Using model interpretation techniques, like feature importance scores and SHAP values, on the best-performing predictive models will give actionable insights into specific churn drivers. This will make it possible to form targeted and practical retention recommendations for the case company.

These hypotheses will be revisited in the discussion of the results to assess how well the findings of this case study align with these literature-based expectations.

2.7 Thesis contribution

The literature review shows that customer churn is an important and complex problem. This is especially true for B2B SaaS companies that use subscriptions. The way churn is defined and its causes can be very different. Churn involves factors like the quality of the product, customer service, price, the value customers feel they get, how much they use the service, costs for switching to another service, competition, and details about individual customers. (Mutanen, 2006; Rautio, 2019; Keaveney, 1995; Ahn et al., 2006; Dass & Jain, 2011; Wangenheim et al., 2017; Apostolov, 2020.)

About the methods used, research first began with qualitative studies. These tried to understand the reasons why customers leave (Keaveney, 1995). Later, research moved towards using quantitative models which use data to predict churn. Traditional methods like logistic regression are still useful because they are quite easy to understand and provide a basic comparison point (Mutanen, 2006; Stucki, 2019). However, recent studies show that machine learning methods often predict churn more accurately. This is especially true

for methods like Random Forest and Gradient Boosting in recent research about B2B SaaS companies (Keramati et al., 2014; Sabbeh, 2018; Dahlén & Mauritzon, 2023; Khare & Arora, 2024.) Churn data usually has many more customers who stay than customers who leave. Because of this data imbalance, it is important to check the models using more specific measures. Examples are precision, recall, F1-score, and AUC. The best measure to use depends on the company's business goals (Fawcett, 2006; Stucki, 2019; Dahlén & Mauritzon, 2023.)

2.7.1 Identified research gaps

Although research in customer churn prediction has advanced, several areas still need more study. A common limitation mentioned in earlier literature is that studies often use datasets that are limited in their size and variety (Stucki, 2019). For example, Dahlén and Maurizon (2023) point out that using a larger dataset with data from more companies is recommended. Broader datasets would offer additional insights into user behaviors and the reasons leading to customer churn (Dahlén & Maurizon, 2023). An overview of the main research approaches and their commonly discussed limitations is presented in Table 6.

Also, while previous research has identified several key churn drivers, such as platform usage and customer tenure (as summarized in Table 5), the focus often remains on general CRM and usage data. Studies by Dahlén and Maurizon (2023) and Rautio (2019) argue that datasets often miss more nuanced factors that could predict churn, such as detailed information about service problems or the quality of customer support. Rautio (2019) added that future studies should investigate additional available features and apply feature selection methods to identify the most relevant ones. Dahlén and Maurizon (2023) and Stucki (2019) agreed and recommended to try to include these key missing variables and behavioral data. Therefore, testing specific company indicators against more general predictors is an important research area.

Additionally, complex methods like Random Forest are often said to perform best. However, some studies admit that they have only looked at a limited number of different machine learning models (Dahlén & Maurizon, 2023; Stucki, 2019). This shows that more research is needed to compare simpler models, like Logistic Regression and Decision Trees, with more advanced methods like Support Vector Machines, Random Forest, and Gradient Boosting.

Finally, several researchers point out a key gap which is that many quantitative churn studies focus more on prediction accuracy than on understanding and explaining the models. This issue, often referred to as the "black box problem" of machine learning, is a frequently cited limitation in the literature (see Table 6). This means they often do not explain clearly why customers

churn, even though this information is very important for creating good retention strategies. (Dahlén & Maurizon, 2023; Stucki, 2019; Khare & Arora, 2024.) Although it is possible to find out which features are most important, this is not always done. Techniques like SHAP analysis have been mentioned as helpful for this (Dahlén & Maurizon, 2023.) Therefore, more research is needed that builds accurate models but also use interpretation methods to explain the results. Besides these main gaps, studies also mention common challenges with data. These include problems with data quality and incorrect data as well as unbalanced datasets where there are many more non-churned customers than churned ones (Dahlén & Maurizon, 2023; Rautio, 2019).

2.7.2 Positioning this study in B2B SaaS churn research

This thesis aims to add to the understanding of customer churn prediction for B2B SaaS companies by looking at specific research gaps found in earlier studies. This study uses a practical case from a B2B SaaS company and its data. It also continues from an earlier qualitative study done at the same company in 2023. The main ways this thesis contributes are explained next.

First, this thesis answers the call for more studies that compare different prediction models in real company situations (Dahlén & Mauritzon, 2023; Rautio, 2019; Stucki, 2019). Some studies (like Keramati et al., 2014; Sabbeh, 2018) have compared different machine learning models for churn in areas like telecom. Other studies (like Dahlén & Mauritzon, 2023) have tested new models like XGBoost for large SaaS companies. However, not much research has directly compared logistic regression along with several newer machine learning models using a company's own special data from a B2B SaaS business operating in the Nordic market. This study intends to fill that gap. It will give clear results from real data about which models work best for predicting churn in this kind of B2B SaaS company.

Second, this study looks at how well company-specific information and detailed customer behavior data can predict churn. This kind of information is often missing from the general datasets used in churn research (Dahlén & Mauritzon, 2023; Rautio, 2019; Stucki, 2019). In particular, this research tests how useful the case company's own Customer Health Score (CHS) metrics are. These scores include information about how much customers use the service, their support contacts, and their satisfaction. By looking at these scores together with more standard predictors, this study offers useful information on how well such company-specific signs can show churn risk. This goes beyond the usual information like customer size or how much they spend.

Also, this thesis focuses on understanding why models make certain predictions. This follows strong advice from recent studies (Dahlén & Mauritzon, 2023; Stucki, 2019; Khare & Arora, 2024). The aim is not just to build models that predict churn well, but also to get useful ideas for action. By using

methods like feature importance analysis and SHAP (SHapley Additive exPlanations) values, this research works to explain why some customers are likely to churn. This helps in making clear and practical suggestions for the case company on how to keep its customers.

To sum up, this thesis aims to add useful academic knowledge about B2B SaaS churn prediction. It does this by comparing different prediction methods, carefully checking the value of unique company-specific data, and focusing on understanding model results for practical use. At the same time, it aims to give the case company real, evidence-based advice that they can use.

3 Research material and methods

This thesis employs a quantitative research approach within a case study design to investigate customer churn prediction at a specific B2B SaaS company, Lime Technologies Finland. The primary aim is to develop, compare, and evaluate the performance of different machine learning models using the company's historical customer data to identify customers at risk of churning. This task is approached as a binary classification problem, where models learn to distinguish between customers who have churned and those who have remained active.

A key aspect of this study is its direct connection to previous internal research conducted by the case company. It serves as a quantitative follow-up to a qualitative study on churn drivers performed by the company in 2023. This connection informed the current research, particularly in the data selection phase. Following the recommendation of involving domain experts in feature selection (Buö & Kjellander, 2014), the initial variables for this study were chosen in collaboration with the company representatives who conducted the earlier qualitative research.

The overall methodology follows a structured process common in applied machine learning projects, as described in the later sections of this chapter. This process begins with understanding and describing the research material, followed by data preprocessing steps including cleaning, transformation, and handling potential issues like class imbalance. Feature engineering techniques were then applied to create potentially more informative predictors. Subsequently, a selection of machine learning models, chosen based on findings in the literature review and their suitability for this type of classification task, were implemented. The performance of these models was evaluated using appropriate metrics suited for churn prediction, and finally, model interpretation methods were employed to understand the key factors driving the predictions.

This structured, data-driven approach aims not only to achieve strong predictive performance but also to generate practical insights that can support Lime Technologies Finland's customer retention efforts, ensuring the findings are both technically grounded and relevant from a business perspective.

3.1 Case company

The case study in this thesis focuses on customer churn prediction within Lime Technologies Finland, which is part of the larger Lime Technologies AB group. Headquartered in Lund, Sweden, Lime Technologies AB is a publicly listed software company specializing in Customer Relationship Management solutions, with operations across the Europe. The company offers a portfolio of products, including adaptable systems like Lime CRM and tools like Lime

Go designed for B2B sales teams, operating primarily under a Software-as-a-Service model where subscription-based recurring revenue is key.

For companies operating under a SaaS model like Lime Technologies, customer retention is crucial as it directly influences revenue stability and profitability. Acquiring new customers is often significantly more costly than retaining existing ones (Stucki, 2019). High churn rates can erode the customer base and negatively impact growth. Research highlights the financial importance of retention. For instance, Dahlén and Mauritzon (2023) note that even a small decrease in churn can lead to substantial improvements in profitability for SaaS businesses. Therefore, understanding and managing customer churn is a critical strategic objective.

While Lime Technologies has strong retention in many markets, this thesis focuses specifically on the Finnish context because the churn rate in Finland showed an upward trend in recent years. Internal company statistics indicated a churn rate of approximately six percent in Finland for 2023. Although similar to some other Nordic markets, this rate presented an opportunity for improvement, aligning with the company's broader goal of strengthening customer loyalty across all regions. There is thus a clear business motivation within Lime Technologies Finland to explore how data-driven prediction methods can support customer retention efforts.

To better understand customer loyalty, the case company conducted a qualitative study in 2023. That study focused on understanding why customers stay by comparing feedback from different groups: active customers, customers who churned within two years, and customers who churned after a longer period. The current quantitative thesis builds upon those qualitative insights by focusing on developing and evaluating predictive models to identify which customers are likely to churn before they leave, using historical data.

The data used in this study is drawn specifically from Lime Technologies Finland's operations and CRM system. The findings and models developed are therefore tailored to the characteristics and dynamics of this specific Finnish B2B SaaS customer base.

3.2 Data description

Building successful prediction models strongly depends on having good quality data that is relevant and properly organized (Rautio, 2019). For churn prediction, data from Customer Relationship Management systems is often very valuable because these systems store important customer information and track interactions over time (Soltani & Navimipour, 2016). This section describes the data used in this thesis study.

3.2.1 Data sources

The dataset for this study came directly from the case company, Lime Technologies Finland's, own CRM system. Using company CRM data is a common practice in churn analysis as it contains details about customer history and interactions (Sabbeh, 2018). The collected data includes customer-specific information covering the period from 2019 to 2024.

The process of extracting the data was done in collaboration with the company's management team. Their experience in analyzing customer behavior and knowledge from earlier internal research helped in selecting the most relevant data available within the organization for understanding churn.

3.2.2 Defining the target variable

To build a predictive model, it is essential to have a clear definition of what is being predicted. In this study, customer churn was defined as the termination of a customer's contract with Lime Technologies. This means a customer was marked as churned if they officially ended their contractual relationship with the company. This definition aligns with how churn is often understood in subscription-based SaaS businesses (Rautio, 2019).

The dataset contained a specific target variable, named churn_y/n, to indicate the status of each customer. This variable is binary, meaning it has two possible values: 1 for a customer who has churned, and 0 for a customer who is still active. The main task for the machine learning models developed in this thesis is to predict this binary outcome based on the other available customer data.

3.2.3 Data variables

Predicting customer churn successfully often requires using different types of data to get a full picture of the customer relationship (Dahlén & Mauritzon, 2023). Following guidance from the case company's internal research and expertise, the initial data for this study originated from two primary datasets extracted from Lime Technologies Finland's CRM system. These datasets, covering the period 2019-2024, captured different temporal aspects of customer information: monthly health score metrics and yearly financial/interaction data.

The first initial dataset focused on the Customer Health Score (CHS), which includes metrics developed internally by Lime Technologies. This dataset contained monthly records for each customer, tracking the ongoing status of the relationship. It was made up of several components measuring different aspects, each contributing a score, typically ranging from 0 to 4, based on specific criteria. The main components included:

- Usage: Reflecting login activity relative to the number of licenses. This component directly measures key aspects of service usage and engagement, which literature consistently identifies as a critical predictor of churn in SaaS environments (see Section 2.3.4; Rautio, 2019; Apostolov, 2020). Low or declining usage can signal disengagement and a reduced perceived value of the service.
- Stakeholders: Based on the number of key contact persons identified at the customer company, adjusted for company size. A higher number of engaged stakeholders can indicate a deeper and more embedded relationship with the B2B customer, potentially relating to higher switching costs and stronger overall relationship tenure aspects (see Sections 2.3.5 and 2.3.7).
- Stickiness: Representing the level of product customization and use of addons. Increased stickiness through customization and addon adoption is hypothesized to elevate switching costs for the customer (see Section 2.3.5), as it makes the service more tailored to their specific needs and harder to replace.
- Relation: Based on the recency of contact with stakeholders, adjusted for company size. This metric aims to capture the health of the ongoing customer service and support experience and interaction frequency (see Section 2.3.2). Regular and recent positive interactions are generally associated with higher customer satisfaction and loyalty (Section 2.3.7).
- NPS: The customer's Net Promoter Score from recent surveys. NPS is a direct measure of overall customer dissatisfaction/loyalty (see Section 2.3.7) and is widely used as an indicator of customer satisfaction and potential churn risk.

This CHS dataset also contained static information about the customer's size, industry it operates in, and estimate of turnover. These variables represent customer characteristics and demographics (see Section 2.3.8), which can influence willingness to switch and how other factors impact the customer experience (Rautio, 2019). In addition, a key company-specific variable, relation_strength, was included from the CRM system. The purpose of this variable is to measure the strength of the customer relationship. If the status of the customer company is marked as "active customer" then relation_strength is set to a base value of 50. The score increases as interaction records are created under the customer or as the customer creates an SOS ticket. A higher relation_strength score is assumed to reflect a more stable relationship. The selection of CHS components, static variables and the relation_strength variable was guided by both the case company's internal expertise and prior qualitative research on their customer base, and their

alignment with the key categories of churn drivers identified in the broader B2B SaaS literature.

Table 8: Customer Health Score dataset

Feature	Definition
registration_no	Unique identifier.
date	Date of the recorded data.
customer_type	Size of the customer (XXS/XS/S/M/L).
total_points	Customer Health Score metric.
usage	Customer Health Score metric.
stakeholders	Customer Health Score metric.
stickiness	Customer Health Score metric.
relation	Customer Health Score metric.
nps	Customer Health Score metric.
total_score	Customer Health Score metric.
unique_logins	Number of unique logins by customer.
lime_crm_licenses_in_use	Number of Lime CRM licenses currently used.
lime_crm_license_count	Total number of Lime CRM licenses allocated.
no_of_employee_company	Number of employees in the customer's company.
turnover	Annual revenue of the customer's company.
relation_strength	Measure for customer relationship strength.
industry_expert	Field of industry the customer operates in.
churn_y/n	Target variable (Yes/No).

The second initial dataset contained yearly records for each customer, summarizing key financial investments and interactions. These variables aimed to capture the economic aspect of the customer relationship and patterns of engagement over a longer timeframe. Before preprocessing, this dataset included features such as:

- Revenue-related outcomes: These variables reflect the direct financial value and investment a customer represents to the company in terms of subscription, consultation, and service agreement revenues. Changes in revenue streams, or consistently low revenue, could be linked to the customer's perceived value of the service or their overall financial health, both of which are discussed as churn drivers (see Section 2.3.3 and Section 2.3.8).
- License rate information: The average license rate can provide insights into the pricing aspect of the customer relationship (see Section 2.3.3). Significant changes in average rates or changes over time might correlate with customer satisfaction or churn if the pricing is not perceived as fair or competitive.

- Customer interaction metrics: The frequency and nature of direct interactions like customer visits and sales calls can be an indicator of the customer service and support experience (see Section 2.3.2) and the overall health of the B2B relationship.

The selection of these yearly financial and interaction variables was also informed by the case company's domain knowledge regarding indicators of customer value and engagement, and their connection to broader themes of perceived value, pricing satisfaction, and relationship strength found in the churn literature (Chapter 2.3).

Table 9: Financial and interaction dataset

Feature	**Definition**
registration_no	Unique identifier.
year	Year of the recorded data.
outcome_subscrip-tion_this_year	Subscription revenue outcome for the current year.
outcome_subscrip-tion_last_year	Subscription revenue outcome for the previous year.
outcome_upfront_this_year	Upfront payment outcome for the current year.
outcome_upfront_last_year	Upfront payment outcome for the previous year.
outcome_consulta-tion_this_year	Consultation service revenue for the current year.
outcome_consulta-tion_last_year	Consultation service revenue for the previous year.
outcome_sa_this_year	Service agreement outcome for the current year.
outcome_sa_last_year	Service agreement outcome for the previous year.
license_rate_this_year	Average license rate for the current year.
license_rate_last_year	Average license rate for the previous year.
outcome_customer_visits	Number of customer visits recorded.
outcome_sales_calls	Number of sales calls made to customers.
first_year	First year the customer relationship started.
last_year	Most recent year of recorded customer activity.

These two initial datasets provided the raw historical information. The target variable was also linked to each customer. This raw data then underwent the preprocessing and feature engineering steps described below to create a single, unified dataset for modeling.

3.2.4 Data privacy and ethical considerations

This study handled customer data carefully, thinking about data privacy and ethics. It follows relevant data protection rules, mainly the General Data Protection Regulation (GDPR), as the case company mostly operates in the European Union. Permission to use anonymized company-level data for the purposes of this master's thesis was obtained from the case company's management.

The data used for this research was only about B2B customer companies. Before any analysis, steps were taken to make sure the data did not identify specific companies. All direct identifiers, like company names or contact details, were left out of the datasets extracted from the case company's CRM system. The only identifying information in the original datasets was the company registration number. This variable was used only for the technical purpose of merging the two primary datasets. After that, the registration number was also removed from the final dataset used for analysis. This made sure that the data used for the research was effectively anonymous at the company level. No personal information about individuals was included in the final dataset.

All research findings and results in this thesis will be reported in a way that combines and anonymizes the information. This makes sure that no single customer company can be identified from the published results. The main ethical consideration in this B2B context is to protect business-sensitive company information, which has been addressed through the anonymization of the data and aggregated reporting.

3.3 Data preprocessing

Raw customer data gathered for churn prediction often requires preparation before it can be effectively used in machine learning models (Rautio, 2019). Therefore, after obtaining the initial monthly CHS and yearly financial datasets, several preprocessing steps were necessary. These steps aimed to improve data quality and consistency within each dataset before feature engineering and merging. Common preprocessing activities include data cleaning, handling missing values, and data transformation (Buö & Kjellander, 2014).

3.3.1 Data cleaning

Data cleaning focuses on fixing errors and inconsistencies to improve data quality (Buö & Kjellander, 2014). In this study, the first cleaning step involved ensuring numerical data was in the correct format. Several features

representing monetary values in the yearly financial dataset needed conversion into a standard numerical format suitable for calculations.

Another important part of cleaning is identifying and removing features that could unfairly influence the model because they leak information about the outcome. During inspection of the initial datasets, features identified as potential sources of data leakage regarding churn status, for example "last_year" and "first_year", were removed before proceeding with modeling.

Furthermore, during the initial data exploration phase of the monthly CHS dataset, it was discovered that the variables "unique_logins" and "lime_crm_licenses_in_use" had a very high proportion of missing values. Due to these data quality issues, these two features were considered unreliable and were removed during the cleaning stage.

3.3.2 Handling missing values

Missing data is a common issue in real-world datasets. Simply removing every customer record with missing information is often not a good strategy, as it can lead to losing valuable data from other fields for that customer (Buö & Kjellander, 2014). A better approach is often to fill the missing values using a reasonable strategy.

In this study, missing values were observed in several numerical variables after the initial cleaning steps. Different strategies were used depending on the nature of the variable. For activity-related fields where missing data indicated no activity occurred, zero imputation was used. For other numerical fields containing missing values, median imputation was applied. The median was chosen as it is less sensitive to outliers compared to the mean.

3.3.3 Data transformation

Data transformation involves converting data into a format that is more suitable for machine learning algorithms (Buö & Kjellander, 2014). Many algorithms cannot directly handle categorical features. In this study, the dataset contained some categorical features, such as customer_type and industry_expert. To prepare these for the machine learning models, one-hot encoding was applied. This technique converts each category within a categorical feature into a new binary feature. This ensures that the algorithms can process this information without assuming any incorrect ordering between the categories. All relevant categorical features in the dataset were transformed using this method.

3.4 Feature engineering and final dataset

Raw, time-series data often needs transformation into a static format for standard classification models. Feature engineering involves creating new, potentially more useful, input features from the existing data, and it is recommended to use domain knowledge when creating new features (Buö & Kjellander, 2014). This section describes how features were engineered from the preprocessed monthly CHS and yearly financial datasets, and how the final dataset for modeling was constructed.

3.4.1 Feature engineering

The preprocessed monthly CHS data contained historical information tracking health score components over time. To capture trends, slope analysis was performed. This involved calculating the linear slope for each key CHS metric over the available monthly history for each customer, resulting in features like slope_analysis_of_total_points.

The preprocessed yearly financial and interaction data contained annual summaries. To create overall customer-level indicators, aggregation techniques were applied. This involved calculating total sums and averages across all available years for each customer. Static features like years_as_customer and months_as_customer were also calculated.

3.4.2 Merged dataset and feature selection

The engineered features derived from the monthly Customer Health Score (CHS) data, the aggregated yearly financial and interaction data, and relevant static customer information were then merged into a single dataset. This final dataset, where each row represents one customer, was used for training and evaluating the machine learning models.

The selection of features for the final dataset was guided by the aim to capture a comprehensive view of the customer relationship, taking into account the insights from both the literature review (Chapter 2.3) and the case company's domain expertise. The features can be broadly categorized as follows:

- Customer Health Score (CHS) indicators and trends: These include both the latest available scores for CHS components and engineered features representing the trend of these scores over time. Capturing trends is crucial, as a declining pattern of customer health or engagement can be a more powerful warning signal of churn risk than a single static value, as it reflects the dynamic nature of customer behavior (as discussed in Section 2.3.4 and Section 2.3.7).

- Static customer characteristics: variables such as customer_type, no_of_employee_company, turnover and industry_expert provide the relevant information about the state of the company. As highlighted in section 2.3.8, company-specific demographic information is often linked to different churn rates, especially in the B2B context.
- Contractual and relationship history: Features like lime_crm_license_count and months_as_customer reflect the scale of the customer's commitment and the length of the relationship (Section 2.3.7). The variable “relation_strength” provides another company-specific measure of this. Longer tenure and larger commitments often correlate with customer satisfaction.
- Aggregated financial and interaction metrics: Summarized yearly data such as total_subscription, total_consultation, avg_license_rate, total_visits, and avg_sales_calls_per_year, along with indicators of recent financial activity like latest_subscription, provide insights into the customer's financial investment, perceived value, and interaction levels with the company. These relate to the discussions on pricing and perceived value (Section 2.3.3) and customer service and support experiences (Section 2.3.2).

Regarding feature selection for the final model, the process was primarily guided by domain expertise from the case company and insights from the literature review (Chapter 2.3) regarding established churn drivers. The feature engineering phase (Section 3.4.1) focused on creating meaningful summaries and trend indicators from the initial datasets. After data cleaning steps which removed features prone to data leakage or with significant quality issues, all remaining engineered and static features in the final merged dataset were included for modeling. Additional feature selection steps were not considered necessary at this stage, as the initial selection and engineering were designed to capture theoretically relevant and company-specific indicators. After these feature engineering and merging steps were completed, the final dataset used for training and evaluating the models consisted of 337 unique customer entries. Each row represented a single customer with a comprehensive set of predictors and the corresponding binary churn outcome. The final dataset features, and their definitions, are presented in Table 10.

Table 10: Final dataset features and their definitions

Feature	**Definition**
customer_type	Size of the customer (XXS/XS/S/M/L).
total_points	Customer Health Score metric.
slope_analysis_of_total_points	Trend of change in total points over time.

usage	Customer Health Score metric.
slope_analysis_of_usage	Trend of usage over time.
stakeholders	Customer Health Score metric.
slope_analysis_of_stakeholders	Trend of stakeholder engagement over time.
stickiness	Customer Health Score metric.
slope_analysis_of_stickiness	Trend in customer stickiness over time.
relation	Customer Health Score metric.
slope_analysis_of_relation	Trend in relationship strength over time.
nps	Customer Health Score metric.
slope_analysis_of_nps	Trend in Net Promoter Score over time.
total_score	Customer Health Score metric.
slope_analysis_of_total_score	Trend in total score over time.
lime_crm_license_count	Total number of Lime CRM licenses allocated.
no_of_employee_company	Number of employees in the customer's company.
turnover	Annual revenue of the customer's company.
relation_strength	Measure for customer relationship strength.
industry_expert	Field of industry the customer operates in.
months_as_customer	Number of months a customer has been active.
years_as_customer	Number of years a customer has been active.
total_subscription	Total subscription revenue from the customer.
total_consultation	Total consultation service revenue from the customer.
total_upfront	Total upfront payments made by the customer.
total_sa	Total service agreements made with the customer.
avg_license_rate	Average cost per license sold to the customer.
total_visits	Total number of visits made to the customer.

total_sales_calls	Total number of sales calls made to the customer.
avg_visits_per_year	Average number of visits per year.
avg_sales_calls_per_year	Average number of sales calls per year.
latest_subscription	Most recent subscription value.
latest_license_rate	Most recent average license cost.
churn_y/n	Target variable (Yes/No).

3.5 Handling imbalanced data

Before training the models on the final merged dataset, steps were taken to address the common challenge of class imbalance in churn prediction (Burez & Van den Poel, 2009; Peddarapu et al., 2022). This imbalance can cause standard machine learning models to perform poorly in identifying the crucial minority class, as they might achieve high overall accuracy simply by predicting everyone as a non-churner. (Kotsiantis et al., 2006.) To address this issue in this study, two main steps were taken. First, when splitting the data into training and testing sets, an 80/20 ratio was used. Importantly, this split was stratified. Stratification ensures that the proportion of churned and non-churned customers in both the training set and the testing set is the same as in the original dataset. This provides a more reliable evaluation of the model's performance on unseen data.

Second, to help the models learn better from the underrepresented churn cases, a data generation technique was employed. The Synthetic Minority Oversampling Technique (SMOTE) was chosen for this purpose, as it is a popular and effective method highlighted by researchers like Kotsiantis et al. (2006) for handling class imbalance. Unlike simple oversampling which duplicates existing minority cases, SMOTE generates new, synthetic examples based on the characteristics of existing churners. This process creates a more balanced class distribution in the training data, which can lead to better model performance in identifying actual churners (Burez & Van den Poel, 2009), while avoiding some pitfalls of simple duplication. SMOTE was applied only to the 80% training dataset after the split. This is important to prevent information from the synthetic samples leaking into the test set, which would give an unrealistic evaluation of the model's true performance.

3.6 Churn prediction models used

As discussed in the literature review, various statistical and machine learning approaches have been used to predict customer churn. The models for this study were chosen based on both the literature and the small dataset size

(N=337). For this reason, very complex models like deep neural networks, which need large amounts of data to perform well and avoid overfitting, were excluded from this study. This approach is consistent with findings from previous B2B SaaS research (e.g., Rautio, 2019). Instead, this study focused on a selection of common and reliable models, ranging from a traditional statistical baseline to powerful ensemble methods that are known to perform well on smaller datasets.

The selection included Logistic Regression as a common baseline, Decision Tree for its interpretability, Support Vector Machine for its ability to handle non-linearities, and two powerful ensemble methods, Random Forest and Gradient Boosting, often found to be top performers in churn prediction tasks (Peddarapu et al., 2022; Dahlén & Mauritzon, 2023). For the machine learning models requiring it (SVM, Decision Tree, Random Forest, and Gradient Boosting), hyperparameter tuning was performed using GridSearchCV with 5-fold cross-validation to optimize their performance. Detailed information on the parameter grids and selected optimal parameters for each model can be found in Appendix A. The following subsections briefly describe each model implemented in this study.

3.6.1 Logistic Regression

Logistic regression is a statistical classification technique commonly used for binary outcomes like churn/no-churn (Sabbeh, 2018). It works by modeling the probability of churn using a logistic function based on the input features (Stucki, 2019). Coefficients are estimated for each feature, indicating their influence on the likelihood of churn, which makes the model relatively straightforward to interpret (Stucki, 2019). Due to its simplicity, interpretability, and history as a benchmark in churn studies (Günther et al., 2014), Logistic Regression was included in this thesis as a baseline model against which more complex methods could be compared. A known limitation is its assumption of a linear relationship between predictors and the outcome, which might not capture all complexities in the data (Günther et al., 2014). This can be particularly challenging in B2B SaaS contexts where customer behavior can be influenced by a wide range of non-linear factors and complex interactions. However, its ability to function with potentially smaller datasets often found in B2B settings, and its ease of interpretation make it a valuable starting point, especially when understanding the overall direction of key drivers is important for business stakeholders.

3.6.2 Support Vector Machine

Support Vector Machines are supervised learning models that aim to find an optimal dividing line that separates the different classes with the largest

possible margin (Çelik & Osmanoğlu, 2019). A key strength of SVM is its ability to handle non-linear relationships by using kernel functions. These functions can map the data into higher dimensions where a linear separation might become possible (Çelik & Osmanoğlu, 2019). SVM has shown strong performance in some churn prediction studies, sometimes outperforming other complex models when tuned properly (Rautio, 2019). SVM was included in this thesis comparison to evaluate its effectiveness in capturing potentially non-linear patterns within the case company's churn data, although it can require careful parameter tuning (Rautio, 2019; Çelik & Osmanoğlu, 2019). While SVMs can be powerful in capturing non-linearities, their application to B2B SaaS data, which may contain numerous categorical features requiring preprocessing, can lead to high-dimensional spaces where model performance and training efficiency need careful consideration. Furthermore, the interpretability of SVM models is generally lower than that of simpler models like logistic regression or decision trees, which can be a drawback in B2B contexts where understanding the reasons behind predictions is often highly valued.

3.6.3 Decision Tree

Decision Tree algorithms create a model that predicts the outcome by learning simple decision rules inferred from the data features. The model structure resembles a flowchart or tree, where data is recursively split into smaller subsets based on feature values until final predictions are reached at the 'leaf' nodes (Çelik & Osmanoğlu, 2019). Decision trees are popular because their structure is intuitive and easy to understand, allowing stakeholders to follow the decision logic (Çelik & Osmanoğlu, 2019). This transparency and the ability to generate rule-based explanations can be particularly appealing in B2B settings for communicating insights to sales or customer success teams. While single decision trees can train quickly and handle different variable types, they have a tendency to overfit the training data if not carefully controlled (Sabbeh, 2018; Stucki, 2019), and their structure can be unstable, meaning small changes in data can lead to different trees, which should be considered when relying on them for critical business decisions. In this study, a single Decision Tree model was implemented both for its interpretability and as a fundamental building block for the Random Forest ensemble method, which addresses some of these stability issues.

3.6.4 Random Forest

Random Forest is an ensemble learning method designed to improve the stability and accuracy of single decision trees by reducing overfitting (Stucki, 2019). It works by constructing a large number of individual decision trees

during training. Each tree is built using a different random sample of the data and considers only a random subset of features at each split point (Fang et al., 2016; Rodriguez et al., 2006). The final prediction is made by aggregating the predictions of all individual trees. This approach often leads to robust models with high predictive accuracy on churn tasks (Mau et al., 2018; Peddarapu et al., 2022). Random Forest is well-suited for typical B2B SaaS churn data, which often includes a mix of numerical and categorical variables and complex, non-linear relationships. Random Forest was included in this thesis as it represents a widely used and often high-performing ensemble technique in churn prediction literature. It also provides measures of feature importance, which helps in interpretation (Dahlén & Mauritzon, 2023; Stucki, 2019).

3.6.5 Gradient Boosting

Gradient Boosting is another powerful ensemble technique that builds models sequentially. Unlike Random Forest where trees are built independently, Gradient Boosting builds trees one after another, with each new tree trying to correct the errors made by the previous ones. This iterative process focuses on the cases that are more difficult to predict (Stucki, 2019). Implementations like XGBoost are highly optimized versions known for their efficiency and accuracy, often performing very well in competitions and practical applications like churn prediction (Celik & Osmanoğlu, 2019). Recent studies on B2B SaaS churn have often found Gradient Boosting methods to yield the best predictive performance (Dahlén & Mauritzon, 2023). The high accuracy and efficiency of Gradient Boosting methods make them attractive for B2B SaaS churn prediction, where identifying at-risk, high-value customers precisely is critical. Therefore, a Gradient Boosting model was included in this thesis comparison to evaluate this powerful ensemble approach on the case company data. Although the model is effective, caution must be taken to avoid overfitting through techniques such as regularization and cross-validation (Dahlén & Mauritzon, 2023). While Gradient Boosting models can be more complex to interpret directly compared to simpler models, techniques like SHAP values (as discussed in Section 2.4.3) can be used to understand the drivers behind their predictions, addressing the need for actionable insights in B2B settings.

3.6.6 Ethical considerations in using predictive models for churn

While predictive churn models can help businesses keep customers, it is also important to think about the ethical questions related to making and using them. These questions are different from data privacy (discussed in Section

3.2.4) and are more about how the models work and how their results are used.

One key ethical worry is the chance of unfairness or bias in the algorithms. If the old data used to train a churn model has existing biases, for example, if some customer groups have received poorer service in the past and churned more, or if customer details accidentally link to protected groups, the model might learn and continue these biases. This could mean the model unfairly marks certain types of customers as high churn risks more often than others. This might not be because of what they actually do or need, but because of these learned biases. This could lead to unfair retention actions, or, on the other hand, some customer groups might not get the attention they need. So, it is important to be aware of the data sources. It is also good to check model predictions for different customer groups, if possible, although a very detailed check for fairness is not part of this specific thesis.

Another thing to consider is how clear and easy to understand churn predictions are, especially when these predictions lead to customers being treated differently. If a customer is contacted with a special offer to keep them, or if their service is changed based on a churn prediction, there can be ethical questions if the reasons are not clear or seem random to the customer. Even if very accurate "black box" models work well, not understanding why a customer is flagged can make it hard to be sure that actions are fair, have good reasons, and truly help the customer relationship. Using models that are easier to understand or using tools to explain models (as discussed in Section 2.4.3), can help reduce this problem by giving information about what causes the predictions.

Also, the reason for using churn predictions and how they are used should be ethically right. The main goal should be to improve customer experience, fix the basic problems causing unhappiness, and offer real value to at-risk customers to encourage them to stay. Using churn predictions only for aggressive or tricky retention tactics that do not truly help the customer, or to give less service to those predicted to churn, could be ethically wrong. It could also harm the company's reputation in the long run.

In this study, the goal is to create models that find at-risk customers. This way, the case company can understand their needs better and possibly offer support or solutions that improve their view of the service's value and their experience. The focus is on using the information for positive customer retention, not for unfair or harmful practices.

Table 11 below summarizes the selected churn prediction models, highlighting their key strengths, weaknesses, and considerations relevant to their use in this B2B SaaS case study.

Table 11: Selected Machine Learning models in this study

Model	Key strengths in B2B SaaS churn context	Key challenges in B2B SaaS churn context	Considerations for the case study
Logistic Regression	-Relatively simple to implement and understand. -Provides a good baseline for comparison. -Can offer some interpretability of general driver direction. -Can work with smaller B2B datasets. -Outputs churn probabilities useful for targeting.	-Assumes linear relationships, may not capture complex B2B SaaS behavior. -Struggles with feature interactions. -Coefficient interpretation difficult with multicollinearity. -Static model may need frequent updates. -Sensitive to class imbalance.	-Included as a benchmark. -Helps to evaluate if more complex models offer significant improvement.
Decision Tree	-Very intuitive and easy for non-technical stakeholders to understand. -Transparent model structure. -Can generate rule-based explanations for B2B teams. -Handles different variable types.	-Prone to overfitting if not controlled. -Can be unstable (small data changes lead to different trees). -May not capture very complex relationships as well as ensembles. -Single trees might not have highest accuracy.	-Implemented for high interpretability and as a foundation before Random Forest. -Useful for initial insights into decision rules.
Support Vector Machine (SVM)	-Effective at handling non-linear relationships with kernels. -Shown strong performance in some churn studies when tuned (Rautio, 2019). -Good for finding optimal class separation.	-Sensitive to kernel/parameter choice (requires tuning). -Interpretability generally low, a drawback in B2B. -Performance affected by preprocessing of categorical B2B data.	-Included to evaluate its effectiveness in capturing non-linear patterns in the case data.

Random Forest (RF)	-Generally high predictive accuracy and robust. -Reduces overfitting compared to single trees. -Handles mixed B2B SaaS data types well. -Provides feature importance insights -Good at capturing complex, non-linear relationships.	-Can be a "black box" to some extent (individual prediction logic hard to trace). -May require more computational resources than simpler models.	-Included as a widely used, high-performing ensemble.
Gradient Boosting	-Often achieves very high predictive accuracy in churn (Dahlén & Mauritzon, 2023). -Provides feature importance insights -Handles various data types and complex relationships.	-Prone to overfitting if not carefully tuned/regularized. -Generally more of a "black box" than RF. -Can be more sensitive to hyperparameters than RF.	-Included for its high performance in previous B2B SaaS studies. -Interpretability via SHAP.

3.7 Model evaluation metrics

After building the churn prediction models, it is crucial to accurately evaluate their performance. Different metrics help to check aspects like overall correctness, the balance of different types of errors, and how well a model identifies customers who are likely to churn. These metrics are often calculated from a confusion matrix, which compares the model's predictions to the actual outcomes (Dahlén & Mauritzon, 2023; Stucki, 2019). As discussed previously (see Section 2.5), choosing the right metrics is very important, especially because churn datasets commonly have a class imbalance, meaning there are many more non-churning customers than churning ones. This section defines the specific metrics used in this thesis to evaluate and compare the implemented models, with a particular focus on their relevance to the case company's B2B SaaS churn problem.

3.7.1 Accuracy

Accuracy measures the overall correctness of the model, meaning the proportion of total predictions that the model got right (Fawcett, 2006).

The formula for accuracy is

$$Accuracy = \frac{correct\ classifications}{total\ classifications} = \frac{(TP + TN)}{(TP + TN + FP + FN)}.$$

While accuracy is easy to understand, it can be a misleading metric for churn prediction because churn datasets are often imbalanced. For example, if very few customers actually churn, a model that predicts no one will churn might show high accuracy. However, such a model would be useless for identifying any at-risk customers (Fawcett, 2006). Therefore, in this study, while accuracy will be calculated for completeness, it is not considered the primary metric for evaluating model performance due to the class imbalance in the case company's data. Greater emphasis will be placed on metrics that better reflect performance on the minority class.

3.7.2 Precision

Precision measures the accuracy of the positive predictions made by the model. In churn prediction, it shows the proportion of customers predicted to churn who actually did churn (Stucki, 2019). A high precision score means that when the model identifies a customer as likely to churn, that prediction is usually correct. This helps to minimize the number of False Positives (FP), which are loyal customers incorrectly flagged as churners.

The formula for precision is

$$Precision = \frac{correctly\ classified\ actual\ positives}{everything\ classified\ as\ positive} = \frac{TP}{(TP + FP)}.$$

High precision is important when the actions taken to keep customers are expensive, as it helps to avoid wasting resources on customers who were not actually going to leave (Stucki, 2019). While important for resource efficiency, in this case study, maximizing precision is considered secondary to ensuring that as many true potential churners as possible are identified, due to the high cost associated with losing an established B2B customer.

3.7.3 Recall

Recall, also known as Sensitivity or True Positive Rate (TPR), measures the model's ability to identify all actual positive instances. In the context of churn, it calculates the proportion of customers who actually churned that the model successfully detected (Stucki, 2019). A high recall score indicates that the model is good at finding most of the customers who are truly at risk of leaving, thereby minimizing the number of False Negatives (FN), which are actual churners that the model failed to identify.

The formula for recall is

$$Recall = \frac{correctly\ classified\ actual\ positives}{all\ actual\ positives} = \frac{TP}{(TP + FN)}.$$

High recall is crucial when the cost of missing a potential churner, and therefore losing that customer, is high. For this case study, achieving a high recall is a primary objective. The financial impact of losing an existing B2B customer, along with the associated long-term relationship value, is typically substantial. Therefore, it is more critical for the case company to identify as many at-risk customers as possible so that proactive retention measures can be considered, even if this leads to some loyal customers being flagged. While there is often a trade-off between precision and recall, the emphasis here is on minimizing missed churn opportunities.

3.7.4 F1-Score

The F1-score combines precision and recall into a single metric by calculating their harmonic mean. This score is particularly useful when evaluating models on imbalanced datasets, such as churn data, because it only gives a high score if both precision and recall are reasonably good (Vafeiadis et al., 2015; Stucki, 2019).

The formula for the F1-score is

$$F1\ score = 2 \cdot \frac{precision\ \cdot recall}{precision + recall}.$$

An F1-score close to 1 indicates strong overall performance in identifying actual churners while also being accurate in those predictions. In this study, while recall is the primary focus, the F1-score will also be reported. It

provides a valuable balanced measure, especially when comparing different models that might have varying trade-offs between precision and recall. A good F1-score would indicate that a model is not only good at finding churners but also does so with reasonable accuracy.

3.7.5 Confusion Matrix

A confusion matrix is a table that provides a detailed breakdown of a classification model's performance. It compares the model's predicted classes (churn/no churn) against the actual classes (Dahlén & Mauritzon, 2023). The matrix shows four key counts:

True Positives (TP): Customers correctly predicted as churners.
False Positives (FP): Customers incorrectly predicted as churners.
True Negatives (TN): Customers correctly predicted as non-churners.
False Negatives (FN): Customers incorrectly predicted as non-churners.

Table 12: Example structure of confusion matrix

	Predicted Churn	**Predicted Stay**
Actual Churn	True Positive	False Negative
Actual Stay	False Positive	True Negative

The confusion matrix is essential because it is used to calculate metrics like precision, recall, and accuracy. In this study, analyzing the confusion matrix for selected models will be important. It will help to directly see the number of False Negatives (missed churners), which is a key concern for the case company, and False Positives (incorrectly flagged loyal customers). This detailed error analysis provides practical insights beyond single summary metrics, helping to understand the real-world implications of a model's predictions (Stucki, 2019; Dahlén & Mauritzon, 2023).

3.7.6 AUC-ROC

The Receiver Operating Characteristic (ROC) curve is a graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied. It plots the True Positive Rate against the False Positive Rate (FPR) at various threshold settings (Fawcett, 2006).

The formula for TPR is

$$TPR = Recall = \frac{TP}{(TP + FN)},$$

and for FPR, the formula is

$$FPR = \frac{FP}{(FP + TN)},$$

which tells how many loyal customers were incorrectly predicted to churn.

The Area Under the ROC Curve (AUC) provides a single number summary of the model's performance across all these thresholds. An AUC of 1.0 represents a perfect model, while an AUC of 0.5 indicates performance no better than random guessing. AUC is a widely used metric in churn prediction because it measures the model's overall ability to distinguish between customers who will churn and those who will not. A key advantage of AUC is that it is not dependent on a specific classification threshold and is less sensitive to class imbalance than accuracy (Fawcett, 2006; Dahlén & Mauritzon, 2023; Stucki, 2019). In this study, AUC will be an important metric for comparing the general discriminative power of the different models, providing a threshold-independent view of their effectiveness in ranking customers by churn risk. A higher AUC generally indicates a better model.

3.8 Model interpretation methods

While evaluation metrics show how well the churn models predict, understanding why they make certain predictions is essential for the case company to get practical insights. As discussed in the literature review (Section 2.4.3), interpretability is important for turning model outputs into effective retention strategies (Stucki, 2019; Khare & Arora, 2024). This is especially true for complex models like Gradient Boosting, where interpretation methods help explain the reasons behind predictions (Dahlén & Mauritzon, 2023). To answer the research question about key churn drivers, this study will use the following methods to interpret the best-performing models identified.

3.8.1 Feature importance

A general understanding of which factors drive model predictions will be obtained through two types of feature importance analysis. First, for the tree-based ensemble models (Random Forest and Gradient Boosting) used in this study, their built-in capability to estimate the relative importance of each input feature will be used. These scores show which features have the strongest overall influence on the models churn predictions. Second, to obtain a more robust and model-agnostic measure of feature importance, Permutation Feature Importance (PFI) will be calculated. PFI assesses the importance of a feature by measuring the decrease in a pre-selected model performance score when the values of that single feature are randomly shuffled in the test dataset. This shuffling breaks the relationship between the feature and the

target variable, and the resulting drop in model performance directly indicates the feature's predictive value. The PFI calculation will be repeated multiple times for each feature to ensure more stable estimates. Furthermore, to assess the statistical significance of the PFI scores, bootstrap p-values will be estimated, helping to distinguish true feature effects from random noise. By comparing the importance scores from both the built-in methods and PFI across the best-performing models, this study aims to identify variables that consistently appear as key drivers of customer churn.

3.8.2 SHAP value analysis

For a more detailed look at how features contribute to predictions for individual customers, SHAP (SHapley Additive exPlanations) values will be used. SHAP analysis explains how much each feature value for a specific customer has pushed the prediction towards or away from churn, compared to the average prediction. This study will follow the approach used by Dahlén and Mauritzon (2023) in their B2B SaaS churn study, where SHAP values helped explain complex model behavior. SHAP summary plots will be generated for the top-performing model. These plots will help understand the overall impact and the direction of each feature's influence (for example, whether higher usage reduces churn risk according to the model). This detailed view will provide more specific insights for the case company than feature importance alone, supporting the development of targeted retention actions.

4 Results

4.1 Performance of machine learning models

This chapter presents the results of the empirical analysis conducted to answer the research questions. It begins by evaluating and comparing the performance of the five implemented churn prediction models: Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, and XGBoost. The models were trained and tested on the case company's historical customer data. Their performance was evaluated using the metrics selected and justified in Section 3.7, chosen for their suitability in handling imbalanced datasets and for addressing the specific objectives of B2B churn prediction in this study.

4.1.1 Key performance metrics

The predictive performance of the selected models was assessed using accuracy, precision, recall, and F1-score. Table 13 summarizes these key performance metrics for each model on the test dataset.

Table 13: Performance metrics of implemented churn prediction models

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.868	0.773	0.81	0.791	0.947
SVM	0.941	0.947	0.857	0.9	0.965
Decision Tree	0.926	0.9	0.857	0.878	0.967
Random Forest	0.941	0.905	0.905	0.905	0.986
XGBoost	0.971	0.913	1	0.955	0.988

Table 13 shows that the machine learning models generally performed better than the Logistic Regression across most metrics. As mentioned earlier, achieving high recall was a primary objective for this study. The XGBoost model achieved perfect performance with the recall of 1.000, successfully identifying all true churn instances. Random Forest also performed very well (recall 0.905), followed by SVM and Decision Tree (both 0.857), while Logistic Regression had a recall of 0.810. The F1-score also showed strong results for the ensemble methods. XGBoost achieved the highest F1-score (0.955), with Random Forest (0.905) and SVM (0.900) also performing well. Regarding precision, SVM achieved the highest score (0.947), with XGBoost, Random Forest, and Decision Tree also showing strong precision. While important for resource efficiency, maximizing precision is considered secondary to recall in this study. Finally, overall accuracy was highest for XGBoost (0.971). However, as discussed in Section 3.7.1 and highlighted by Fawcett

(2006), accuracy alone is not the primary indicator for this imbalanced dataset.

4.1.2 Detailed analysis with confusion matrix

While the summary metrics in the previous section give a good overview of model performance, looking closely at the types of errors each model makes gives more useful information for practice. Table 14 presents these confusion matrices for all models used.

Table 14: Confusion matrices for implemented churn prediction models

Model	True Negative (TN)	False Positive (FP)	False Negative (FN)	True Positive (TP)
Logistic Regression	42	5	4	17
SVM	46	1	3	18
Decision Tree	45	2	3	18
Random Forest	45	2	2	19
XGBoost	45	2	0	21

A key concern for the case company, and a primary focus as outlined in Section 3.7.3, is minimizing False Negatives (FN). The XGBoost model achieved an excellent result with zero False Negatives. Random Forest also performed very well with only two FNs. The SVM model showed the best performance in minimizing False Positives (FP). XGBoost, Random Forest, and Decision Tree each had two FPs, while Logistic Regression had five. These counts directly explain the previously observed recall and precision scores. This detailed error analysis highlights that while XGBoost is excellent at identifying all potential churners, SVM is the most precise in its positive predictions, illustrating the practical trade-offs important for the case company's resource allocation.

4.1.3 ROC curve and AUC analysis

To further assess the model’s ability to distinguish between classes an ROC curve analysis was performed. The Area Under the ROC Curve (AUC) scores, previously presented in Table 13, indicated that XGBoost (AUC = 0.988) and Random Forest (AUC = 0.986) achieved the highest values. Figure 4 illustrates the ROC curves for all five models.

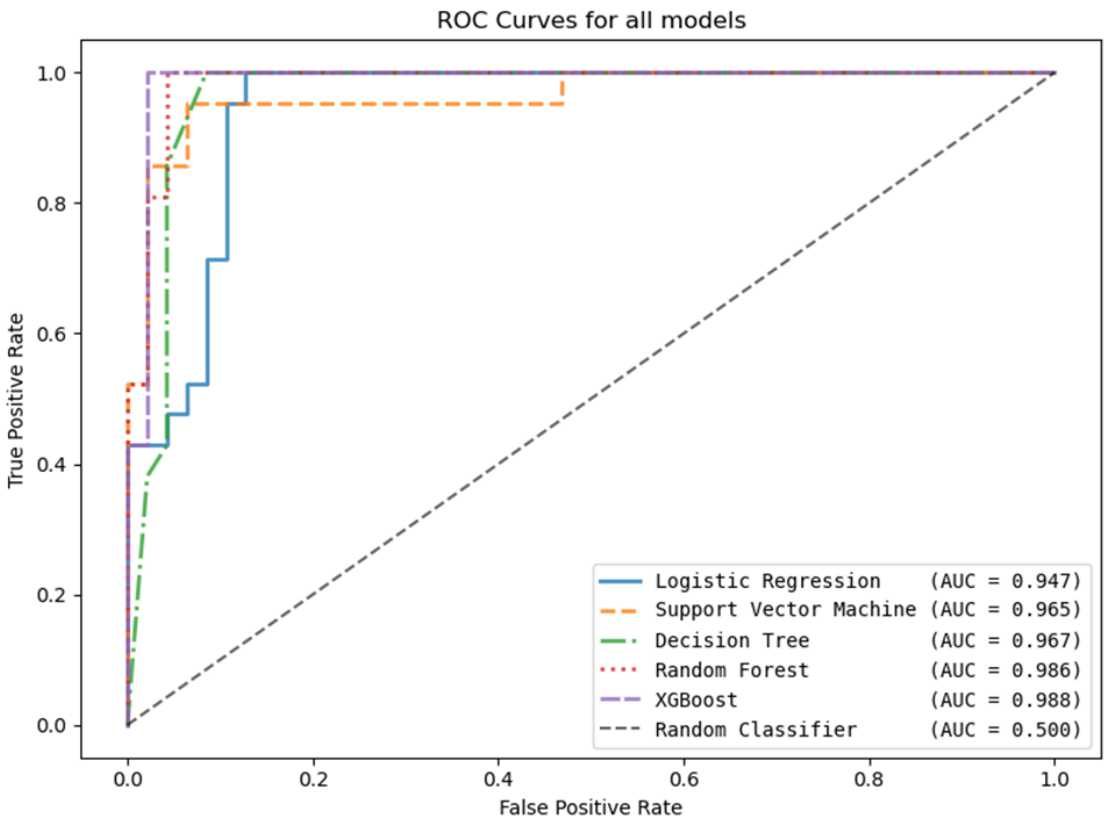

Figure 4: ROC curve for all tested models

As shown in Figure 4, the ROC curves for XGBoost and Random Forest are closest to the top-left corner. The high AUC values, particularly for XGBoost and Random Forest, suggest strong general effectiveness in ranking customers by their likelihood of churn.

4.2 Feature importance analysis

This section presents the findings on which factors the best-performing models, XGBoost and Random Forest, identified as most important for predicting customer churn, addressing RQ2. As described in Section 3.8.1, feature importance was evaluated using both the models internal importance metrics and Permutation Feature Importance (PFI) based on ROC AUC decrease with bootstrap p-values.

4.2.1 Built-in feature importance from models

Tree-based ensemble models, such as XGBoost and Random Forest, provide internal methods to estimate the importance of each feature. These built-in metrics offer an initial overview of which features the models themselves considered most influential during the training phase. In this study, we used

the standard importance metrics. XGBoost's gain measures the total reduction in loss attributed to each split made on a feature, while Random Forest's mean decrease in impurity (MDI) calculates the average reduction in Gini impurity that a feature provides across all trees in the ensemble.

Figure 5 presents the top 10 features as ranked by XGBoost's internal importance metric. Based on this, relation_strength was clearly the most dominant feature. Other features that the XGBoost model considered highly influential included years_as_customer and total_points.

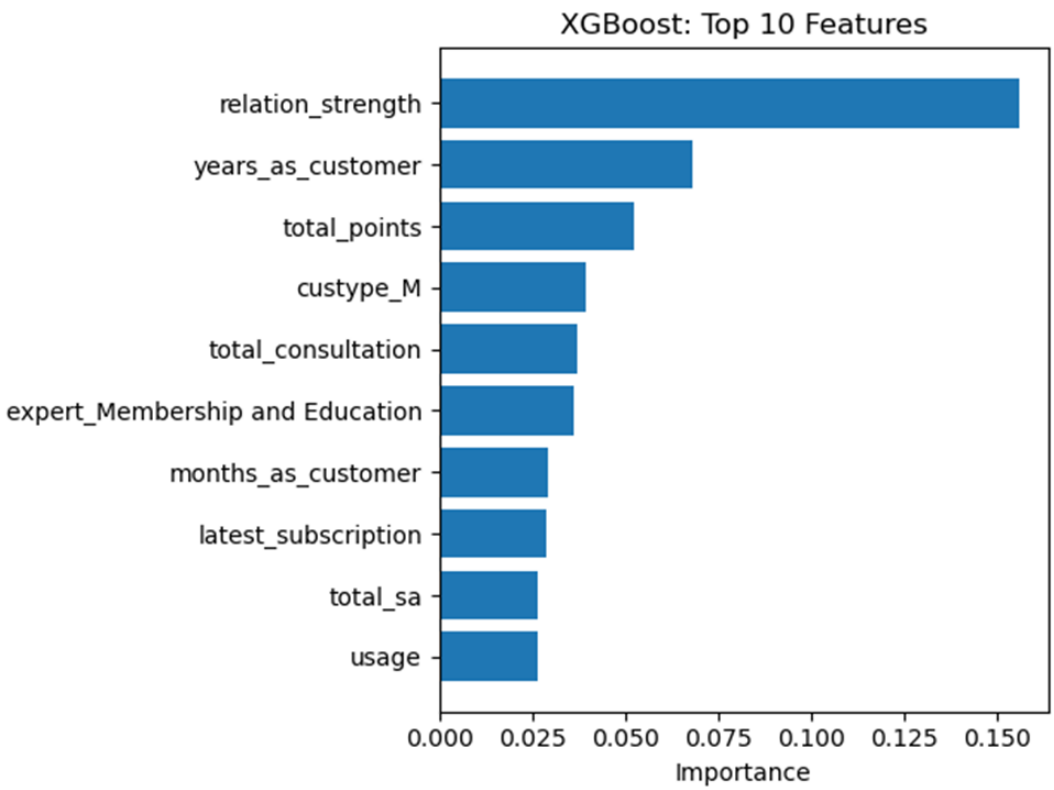

Figure 5: Top 10 Features – XGBoost

Similarly, Figure 6 shows the top 10 features based on the Random Forest model's internal Gini importance. For this model, relation_strength also emerged as a very important feature. The ranking of other features, such as months_as_customer and stickiness, differed from the XGBoost model's internal assessment.

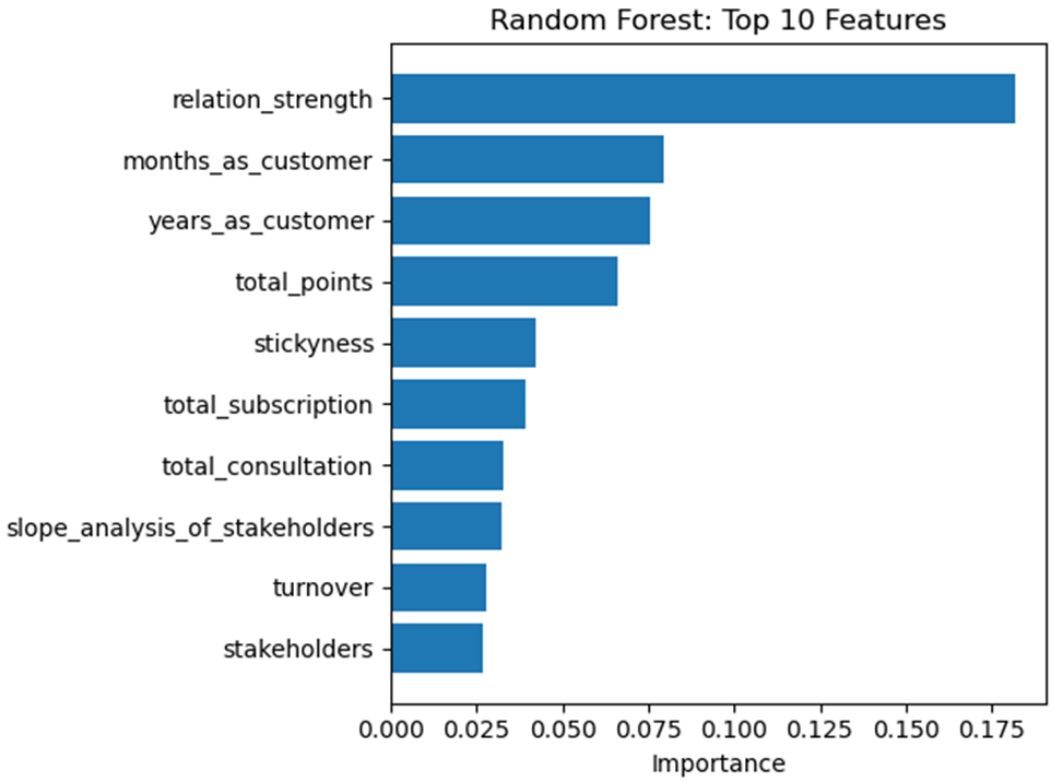

Figure 6: Top 10 Features – Random Forest

While these built-in importance scores provide a first look at feature relevance from the models' internal logic, Permutation Feature Importance (PFI) was also used. PFI offers a different and often more direct way to assess how much each feature contributes to the actual predictive performance of the models on unseen test data, as will be discussed in the next section.

4.2.2 Permutation feature importance (PFI) results

To gain a more direct measure of each feature's contribution to predictive performance on the test set, Permutation Feature Importance (PFI) was calculated. PFI was setup to assess importance based on the decrease in the ROC AUC score. The statistical significance of these importance scores was evaluated using bootstrap p-values.

The PFI results for the XGBoost model are presented in Figure 7. The analysis strongly confirmed relation_strength as the most critical predictor. Shuffling this feature caused a mean decrease of approximately 0.217 in the ROC AUC score, and this impact was found to be highly statistically significant (p-value ≈ 0.002), as indicated by its distinct color in the figure and the displayed p-value. This highlights that the model's ability to distinguish between churning and non-churning customers heavily relies on this particular feature. The feature avg_license_rate showed the second-highest PFI score

with a mean ROC AUC decrease of around 0.009. However, its associated p-value (≈ 0.120) was above conventional significance levels ($p < 0.05$). As seen in Figure 7, the remaining features in the top rankings, such as slope_analysis_of_stakeholders and total_points, had considerably smaller PFI values, and their p-values did not indicate statistically significant contributions to the model's ROC AUC performance in this analysis.

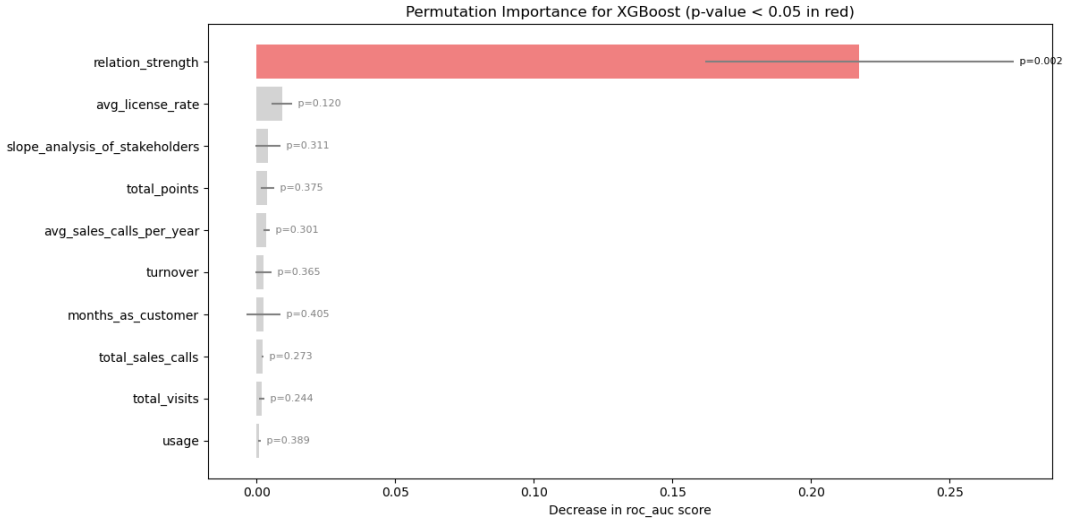

Figure 7: Permutation feature importance for XGBoost (p-value < 0.05 in red)

For the Random Forest model, the PFI results are displayed in Figure 8. Consistent with the XGBoost model, relation_strength was also identified by PFI as the most important and statistically significant feature, with a mean ROC AUC decrease of approximately 0.085 (p-value ≈ 0.004). The feature avg_license_rate again ranked second in importance (mean decrease: ≈ 0.007). For this model, its p-value was approximately 0.074. While this does not meet the strict $p < 0.05$ threshold for significance, it is notably lower than for many other features and might suggest a borderline or weak effect. Other features in the Random Forest's PFI ranking, such as slope_analysis_of_stakeholders and total_points, generally had smaller PFI scores and p-values that did not provide strong evidence of statistical significance for their individual contributions to ROC AUC, as shown in Figure 8.

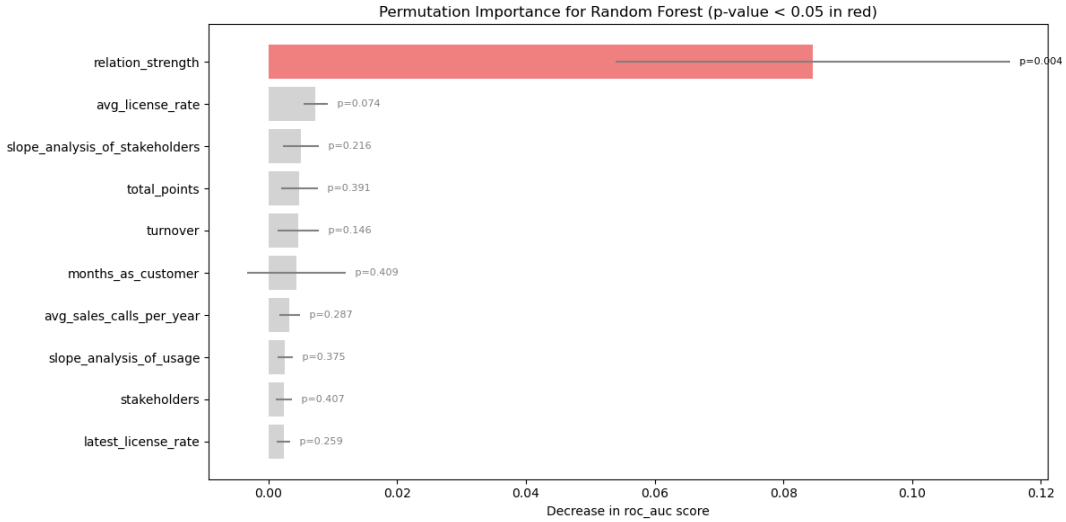

Figure 8: Permutation feature importance for Random Forest (p-value < 0.05 in red)

4.2.3 Synthesis of feature importance findings

Comparing the results from the models internal feature importance measures (Section 4.2.1) with the Permutation Feature Importance (PFI) analysis (Section 4.2.2) gives a broader view of the key churn drivers. A clear finding from all methods and for both the XGBoost and Random Forest models was the great importance of relation_strength. The PFI analysis, with its p-values, further confirmed this, showing relation_strength to be a very statistically significant predictor by a large amount.

While internal importance measures pointed to features like years_as_customer (for XGBoost), the PFI analysis gave a slightly different, and perhaps more direct, view of how features affect prediction accuracy on new data. Notably, avg_license_rate came out as the second most impactful feature in the PFI rankings for both models. This was especially interesting for the XGBoost model, because avg_license_rate was not in its top 10 list based on internal importance scores. This suggests that while avg_license_rate might not be used often for splits inside the XGBoost trees, its presence and correct values are more important for correct churn prediction on the test set than some other features shown by internal measures. For Random Forest, the importance of avg_license_rate was more similar between its internal importance and PFI results, though PFI gave stronger (but still not strictly $p < 0.05$) statistical support for its role.

Based mainly on the PFI results, the most important drivers of churn in this B2B SaaS case study seem to be strongly linked to:

- The strength and nature of the customer relationship, mostly shown by relation_strength.

- Pricing or contract-related elements, with avg_license_rate being the clearest, though its statistical significance in PFI was not as certain as relation_strength.

Other individual features, while appearing in top lists of internal importance or having small PFI values, did not show strong, separate statistical significance in the PFI analysis after considering the main effect of relation_strength. This suggests that while they might play a part, their own unique addition to improving the ROC AUC score, when shuffled individually, was small in these models.

These findings give strong support for the parts of Hypothesis 2 (H2) related to the importance of Relationship strength & trust and some support for Pricing & contractual factors. For other factor groups mentioned in H2, such as those directly showing product-specific engagement or detailed service quality aspects, the PFI analysis did not give strong statistical proof of their individual power to predict ROC AUC in this study. It is possible these factors have a more indirect effect, through interactions with other variables, or that their impact is already largely included in the relation_strength measure in this particular dataset and how the models were built. While PFI shows which features are important for prediction by measuring their impact on overall model performance, it does not fully explain how or in which direction these features affect predictions for individual customers. To get these deeper insights, the SHAP analysis in Section 4.3 will be used next.

4.3 Insights from model interpretability

Section 4.2 showed which features were most important for the prediction models. However, to get useful ideas for action, it is important to understand how these features affect the churn predictions. This section uses SHAP values to look deeper into the best performing model, XGBoost. This helps to understand the key reasons for churn in the case company's customer data. This information helps answer Research Question 2 more fully and gives a base for the practical recommendations for Research Question 3.

4.3.1 Overall feature impact using SHAP summary plot

Figure 9 shows the SHAP summary plot for the top 10 features from the XGBoost model. This plot ranks features by how much they generally impact the model's predictions. It also shows if high or low values of a feature usually lead to a higher or lower chance of churn. Each dot in the plot is one customer from the test set. The color of the dot shows if the feature's value for that customer was high (red) or low (blue). The dot's place on the x-axis shows the SHAP value. This SHAP value is how much that feature's value changed

the churn prediction for that specific customer. Positive SHAP values mean the prediction moved towards churn. Negative SHAP values mean the prediction moved away from churn.

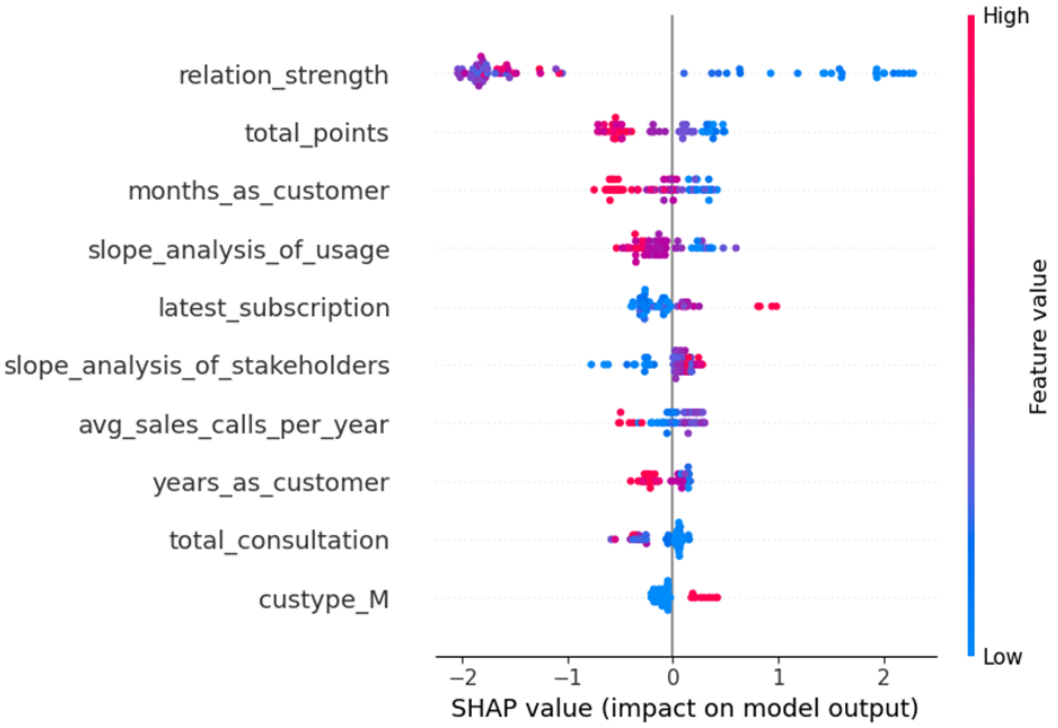

Figure 9: SHAP Summary Plot for Top 10 Features from XGBoost Model

Figure 9 clearly shows that relation_strength has the biggest impact. High values of relation_strength have very negative SHAP values. This means they greatly lower the predicted chance of churn. Low values of relation_strength have high positive SHAP values. This means they strongly increase the churn prediction. This suggests that a strong relationship with the customer is very powerful in stopping churn.

Other important features like total_points and months_as_customer also show clear patterns. Higher total_points and more months_as_customer usually lead to negative SHAP values. Lower scores and shorter times as a customer push towards a positive SHAP value. The plot also shows how features like slope_analysis_of_usage, latest_subscription, and custype_M affect the model's predictions.

4.3.2 How specific features affect predictions

To understand better how the value of a specific feature changes its impact on the churn prediction, SHAP dependence plots were made for the top three most important features from the XGBoost model. Figure 10 is the SHAP dependence plot for relation_strength. When the relation_strength value goes

up, its SHAP value tends to go down quickly from positive to negative. This confirms that as the relationship with a customer gets stronger, the model predicts a much lower chance of churn. This feature strongly pushes customers with very low relation_strength values towards a churn prediction.

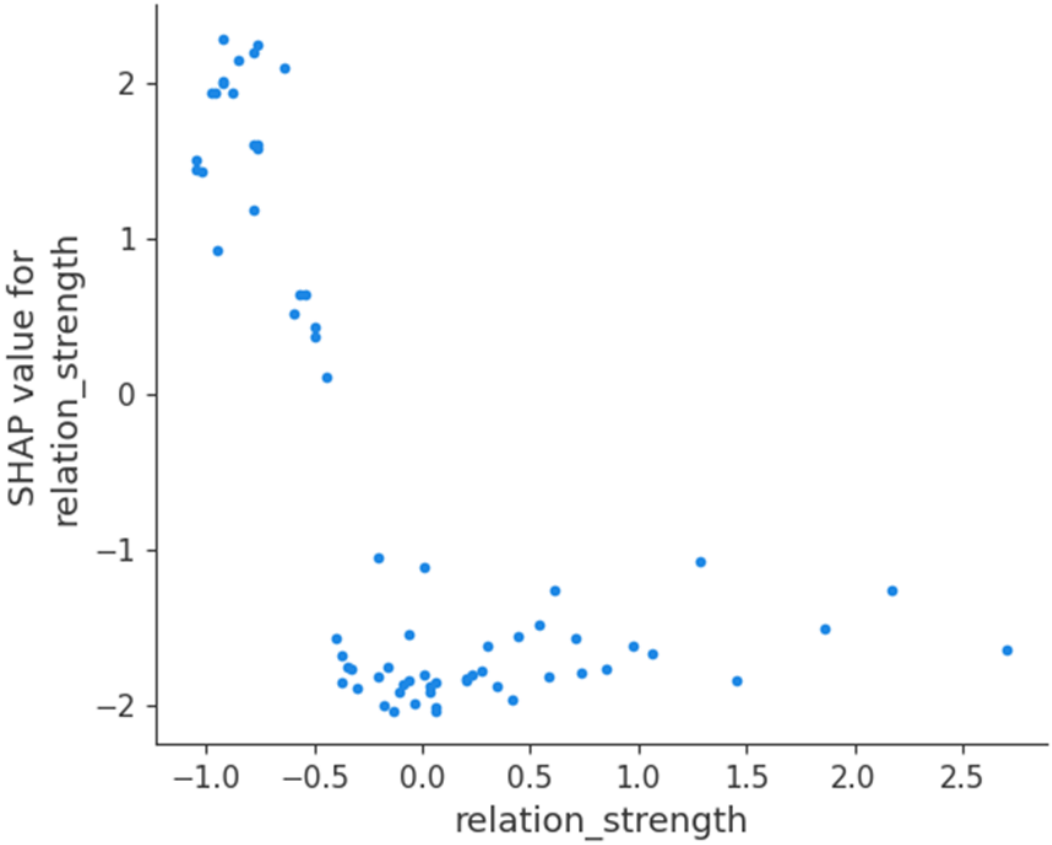

Figure 10: SHAP Dependence Plot for relation_strength

The dependence plot for total_points (Customer Health Score) is in Figure 11. This plot shows that higher total_points values connect with negative SHAP values, which lowers the churn risk. Customers with lower total_points, meaning lower overall health or engagement, have positive SHAP values. This increases their predicted chance of churn. This fits with the idea that healthier, more engaged customers are less likely to leave.

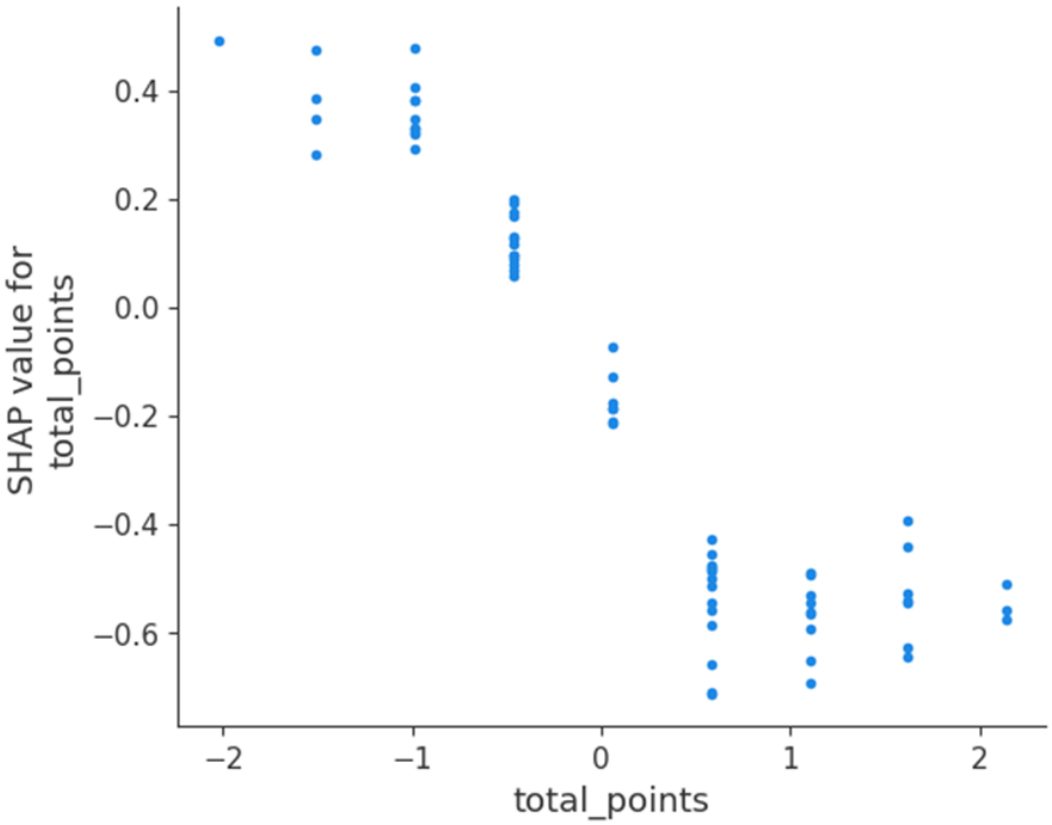

Figure 11: SHAP Dependence Plot for total_points

Figure 12 shows the SHAP dependence plot for months_as_customer. The plot suggests that customers who have been with the company for a shorter time usually have higher SHAP values. This means an increased churn risk. As the number of months as a customer goes up, the SHAP values tend to go down. This suggests that the model predicts long-time customers are less likely to churn. This shows it is important to support newer customers well.

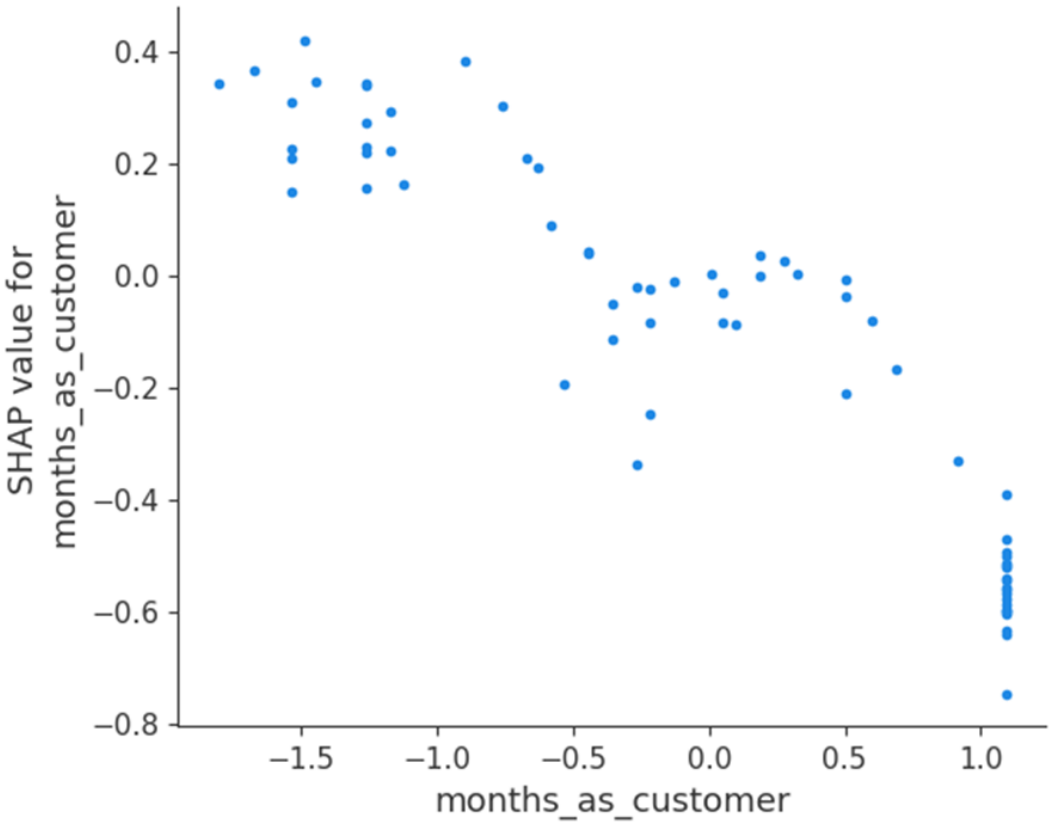

Figure 12: SHAP Dependence Plot for months_as_customer

These SHAP analyses give a more detailed understanding of how the most important features affect the XGBoost model's churn predictions. The findings confirm that things related to the strength of the customer relationship, overall customer health and engagement, and how long they have been a customer are key reasons for churn in this B2B SaaS case. This supports Hypothesis 3, which suggested that model interpretation tools would give useful insights into specific churn drivers. This detailed understanding is important for making targeted and practical retention recommendations for the case company. These will be discussed more in Chapter 5.

5 Discussion

This chapter discusses the empirical findings presented in Chapter 4, interpreting their significance in relation to the research questions and the existing body of literature. The primary objective is to move from reporting results to explaining their meaning and implications. The chapter is structured to first summarize the principal findings, then interpret them in a broader academic context, discuss their theoretical and managerial implications, acknowledge the study's limitations, and finally suggest avenues for future research.

5.1 Key findings

The empirical analysis conducted in this thesis provided several key findings that directly address the three research questions guiding the study. In response to the first research question (RQ1), which aimed to identify the most effective predictive modeling techniques, the findings showed clear differences in model performance. Ensemble-based machine learning methods performed significantly better than the traditional logistic regression baseline. Notably, the XGBoost model stood out as the top performer, achieving an exceptionally high recall of 1.0, which successfully fulfilled the primary business objective of identifying all potential churners. Furthermore, it recorded the highest F1-score (0.955) and Area Under the ROC Curve (AUC) score (0.988), indicating both a strong balance of precision and recall, and an excellent overall ability to separate churners from non-churners. The Random Forest model also performed strongly. These results provide strong empirical support for Hypothesis H1, which suggested that ensemble models would offer higher predictive performance in this B2B SaaS context.

Addressing the second research question (RQ2) regarding the key drivers of customer churn, the analysis consistently and clearly identified the company-specific composite metric, relation_strength, as the most important predictor. Both Permutation Feature Importance (PFI) analysis, which confirmed its statistical significance ($p < 0.05$), and SHAP analysis highlighted its significant impact on model predictions. Specifically, low values of relation_strength were the single most powerful factor driving predictions toward churn. Other significant, although secondary, drivers included the duration of the customer relationship (months_as_customer and years_as_customer) and the overall customer health score (total_points). SHAP analyses confirmed that longer tenures and higher health scores were strongly associated with a lower risk of churn.

Finally, the findings provided a clear path to answering the third research question (RQ3), which focused on creating practical recommendations for customer retention. The successful application of model interpretation

techniques, particularly SHAP value analysis, translated the predictive power of the "black box" XGBoost model into transparent and actionable insights. By visualizing the exact impact and direction of key drivers like relation_strength and months_as_customer, the analysis provided the necessary evidence-based foundation to develop targeted retention strategies. This confirmed Hypothesis H3, which suggested that model interpretation techniques would provide such actionable insights, thereby bridging the gap between prediction and practical application.

5.2 Findings in the context of existing literature

The results summarized in the previous section offer important insights into customer churn prediction in a B2B SaaS environment. When we compare these findings to existing literature, they not only support several known theories but also provide new, specific views on the topic. The most important finding of this study is the key role of the relation_strength variable. This finding strongly supports the Commitment-Trust Theory (Morgan & Hunt, 1994), which states that commitment and trust are very important for successful and lasting business relationships. In this study, the relation_strength metric, which combines interaction history and support contacts, can be seen as a practical measure of these theoretical ideas. The variable's strong influence as a predictor confirms that a strong, active relationship is essential for preventing churn. This also addresses the suggestion from previous research (e.g., Rautio, 2019; Stucki, 2019) that studies should include more detailed, company-specific behavioral data that goes beyond standard usage metrics.

The better performance of the XGBoost and Random Forest models over Logistic Regression aligns with the conclusions of many recent churn studies (e.g., Dahlén & Mauritzon, 2023; Khare & Arora, 2024). The reason for this is likely their ability to capture complex, non-linear relationships within the data. For example, the negative impact of a customer having a short tenure might be much more serious if they also have a low total_points score. Ensemble models are designed to identify such complex patterns, whereas linear models like logistic regression cannot. This confirms that the factors driving B2B SaaS churn are not simple and isolated but are instead connected to each other.

It is also interesting to consider the factors that were not ranked as top predictors in the PFI analysis, such as many of the individual customer health score components. This does not necessarily mean they are unimportant. Instead, it is likely that their individual impact is already captured within the more comprehensive relation_strength score. This suggests that the overall quality of the customer relationship is a more powerful predictor than any single, isolated signal of customer behavior. While a drop in product usage is

an important flag, this study suggests it is the overall pattern of communication and connection that ultimately determines customer loyalty.

Finally, the findings provide clear empirical support for the conceptual model developed in Chapter 2 (Figure 3). The strong performance of relation_strength directly confirms that the “Relationship Strength & Trust” component is a central element influencing the likelihood of churn. The secondary importance of avg_license_rate also supports the “Pricing & Contractual Factors” component. The results therefore confirm the core parts of Hypothesis H2, showing that these specific elements are indeed important indicators of customer churn in the case company's B2B SaaS context.

5.3 Implications of the study

Beyond interpreting the results, it is important to consider what they mean for both academic research and business practice. This section outlines the theoretical contributions of the study to the field of churn prediction. It also discusses the practical, managerial implications for the case company and similar B2B SaaS businesses.

5.3.1 Theoretical implications

This study contributes to the academic literature on customer churn in several ways. First, this research provides a direct comparison of several machine learning models using a real-world dataset from a B2B SaaS company in the Nordic market. By testing models from simple logistic regression to more complex ensembles like XGBoost, the study addresses the suggestion from recent literature (e.g., Dahlén & Mauritzon, 2023) for more head-to-head comparisons in specific business contexts. Second, the study shows the high value of using a company-specific, composite metric like relation_strength. While much of the existing research focuses on general variables like product usage or company size, this study’s finding that an internal relationship metric was the most important predictor suggests that future academic models could be improved. Researchers should consider how to include measures of overall relationship quality, not just simple behavioral data. Finally, this research offers a practical example of how to combine prediction with explanation. By using a high performing "black box" model (XGBoost) and then explaining its predictions with SHAP analysis, the study addresses the common challenge of model interpretability. This provides a useful template for other researchers who want to build accurate models that also offer clear, understandable insights.

5.3.2 Managerial implications

To meet the objective of Research Question 3, this section translates the analytical findings into practical, evidence-based recommendations for the case company and similar B2B SaaS businesses. The findings strongly suggest that the case company should place a strategic focus on proactively managing relationship health. This study's most clear result was the predictive power of the relation_strength metric, which gathers points from customer interactions and support tickets. Because a low score indicates a lack of engagement and a higher churn risk, managers should consider using it as a central key performance indicator (KPI) for customer health. Instead of using it as a lagging indicator, it could form the basis of a proactive retention program. For instance, companies can establish internal health benchmarks. Customers who fail to gain new interaction points over a certain period could be automatically flagged for a proactive check-in by their Customer Success Manager. This approach would shift retention efforts from a reactive to a proactive model, allowing the company to systematically identify and engage with passive or disengaged customers before they become a critical churn risk, thereby preventing revenue loss.

Furthermore, the study highlights the value of integrating predictive analytics into daily customer success operations. The XGBoost model demonstrated its ability to reliably identify customers with a high probability of churning. To make this predictive power actionable, the model's output could be put into practice as a tool, such as a prioritized "at-risk dashboard" for Customer Success Managers (CSMs). Such a tool would not only rank customers by their churn score but could also display the key factors influencing that score for each customer. This would give CSMs data-driven insights, helping them understand not just who is at risk, but also why. This enables more targeted conversations and ensures that retention efforts are consistently focused on the accounts with the highest financial risk, moving beyond intuition-based decision-making.

Finally, the results provide a clear, evidence-based rationale for enhancing the early-stage customer journey. The analysis showed that customers with a shorter tenure are more likely to churn, indicating that the initial phase of the customer lifecycle is the most critical. This suggests that B2B SaaS companies should view their onboarding process as more than just technical training. It is a foundational period for relationship building. For example, implementing structured value check-ins at key milestones, such as 30 or 90 days post-onboarding, could be highly beneficial. The goal of these interactions would be to move beyond asking "how is the software?" and instead focus on whether the customer is achieving the business value they expected. By ensuring new customers feel supported and see a return on their investment quickly, the company can build a stronger foundation for a long-term,

profitable relationship and mitigate the high risk connected to early-stage churn.

5.4 Limitations of the study

To provide a balanced view of this research, it is important to acknowledge its limitations. These limitations help to define the context in which the findings should be understood and offer a foundation for future improvements in this area of study.

First, the primary limitation of this research is related to its generalizability. As a single-case study focused on one company in the Finnish market, the findings may not be directly applicable to other companies, industries, or geographical regions. The exceptional importance of the relation_strength metric is particularly context specific. Other companies would need to develop and test their own similar metrics to see if the same result applies.

Second, there are limitations connected to the data and variables used. The most important predictor, relation_strength, is an internal metric of the case company. While this provided a powerful insight, the exact calculation method is not public. This limits the ability for other researchers to replicate these findings precisely. Additionally, the dataset did not include information on external factors, such as competitor activities or sudden changes in a customer's business, which are known to influence churn but could not be measured in this study.

Finally, some methodological choices also have limitations. This study used the SMOTE technique to address the class imbalance in the training data. While this is a common and effective method for helping the model learn from the minority class, it involves creating synthetic data points. There is a theoretical risk that this could cause the model to overfit to specific characteristics of the synthetic samples. Although the model performed well on the unseen test data, this risk is important to acknowledge.

5.5 Recommendations for future research

Based on the findings and limitations of this study, several areas for future research can be identified. These suggestions aim to build upon the insights from this thesis and to address the questions that remain unanswered.

First, a valuable next step would be to investigate the relation_strength metric further. This study was limited to using the metric as a single input. Future research could aim to break it down. By analyzing its separate parts, such as the frequency of support contacts, the number of key stakeholders, or the time since the last interaction, researchers could identify which specific actions are most critical for building a strong customer relationship. This would produce insights that are more useful for a wider range of B2B SaaS companies.

Second, future research could add statistical testing to the comparison of model performance. While this study showed clear differences in performance metrics like AUC and F1-score, it did not formally test if these differences were statistically significant. Future studies could use statistical tests, such as a paired t-test on cross-validation results or the Wilcoxon signed-rank test suggested by Demšar (2006). This would provide stronger evidence about whether the better performance of an ensemble model like XGBoost over a baseline like Logistic Regression is a stable and statistically meaningful finding.

Third, future studies could explore more time-based approaches to churn prediction. This study used a static model that predicts if a customer will churn. A more advanced approach, like survival analysis, could be used to predict when a customer is most likely to churn. This would provide companies with a time-based risk assessment, helping them to time their retention efforts even more effectively.

Fourth, to significantly increase the statistical power and generalizability of the predictive models, future work should aim to utilize a larger dataset. A clear managerial recommendation for the case company is to expand the scope of the analysis to include customer data from the entire Lime Technologies group, not just the Finnish market. A larger and more diverse dataset would likely produce more robust models that are less prone to overfitting on the specifics of a single market, and it could help uncover more universal churn drivers applicable across the organization.

Finally, there is a good opportunity for a qualitative follow-up study. While this quantitative research identified what factors are important, a qualitative study could explore the why behind them. Conducting in-depth interviews with customers who have both high and low relation_strength scores could provide rich, story-based data. This would help to understand what a "strong relationship" means in practice from the customer's perspective, adding an important layer of understanding to the statistical findings of this thesis.

5.6 Conclusion

The purpose of this thesis was to identify effective predictive methods and the key drivers of customer churn in a B2B SaaS company. By developing and comparing several machine learning models, the study found that modern ensemble methods, especially XGBoost, can predict at-risk customers with very high accuracy. The most important conclusion of this research is that the overall strength and quality of the customer relationship is the most critical factor in preventing customer churn. This was shown by the strong predictive power of relation_strength, a company-specific metric that measures customer engagement and interaction. This finding suggests that for B2B SaaS companies, long-term customer loyalty depends less on single product features and more on the ability to build and keep a strong, trust-based

relationship. In practice, these findings provide a clear, evidence-based direction for retention strategies. By using predictive models to identify at-risk customers and focusing on strengthening relationship health through proactive engagement, companies can effectively reduce churn and support long-term growth. In short, this study shows that the path to customer retention in the B2B SaaS world is built on data-driven insights and a deep commitment to the customer relationship.

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Appendix A: Hyperparameter details for predictive models

GridSearchCV with 5-fold cross-validation was used to identify the parameter combinations that give the best performance on the training data.

Logistic Regression

The Logistic Regression model was implemented using its default parameters as provided by the scikit-learn library.

Support Vector Machine (SVM)

Parameter grid searched:

C: [0.1, 1, 10]
kernel: ['linear', 'rbf']
gamma: ['scale', 'auto']

Optimal parameters found:

C: 10
gamma: 'scale'
kernel: 'rbf'

Decision Tree

Parameter grid searched:

max_depth: [None, 3, 5, 7, 10]
min_samples_leaf: [1, 2, 5, 10]
criterion: ['gini', 'entropy']

Optimal parameters found:

criterion: 'entropy'
max_depth: 5
min_samples_leaf: 10

Random Forest

Parameter grid searched:

n_estimators: [100, 300, 500]
max_depth: [None, 5, 10, 20]
max_features: ['sqrt', 'log2', None]
min_samples_leaf: [1, 2, 5]

Optimal parameters found:

max_depth: None
max_features: 'sqrt'
min_samples_leaf: 1
n_estimators: 500

XGBoost (Gradient Boosting)

Parameter grid searched:

n_estimators: [100, 300, 500]
max_depth: [3, 5, 7]
learning_rate: [0.01, 0.1, 0.2]
subsample: [0.8, 1.0]
colsample_bytree: [0.8, 1.0]

Optimal parameters found:

colsample_bytree: 0.8
learning_rate: 0.01
max_depth: 7
n_estimators: 500
subsample: 0.8